# Supplementary Material: On Latent Predictive Residuals and Automated Threshold Calibration for Multivariate Time-Series Anomaly Detection

## S-1. Proofs of Propositions

*Proof of Proposition 1.* **(Operator Non-Triviality and Barrier Behavior).** (i) At $\mathbf{K} = \mathbf{0}$, under the boundary convention assigning zero spectral entropy $\mathcal{H}_{\text{spectral}}(\mathbf{0}) \equiv 0$, the deficiency attains its upper bound $\Delta\mathcal{H}(\mathbf{0}) = \log D$. The implemented barrier is saturated at its floor: $r_{\text{safe}} = \epsilon$ is constant on a neighborhood of $\mathbf{0}$, so $\Phi_{\text{energy}}$ contributes zero gradient there, while the prediction term has gradient $\nabla_{\mathbf{K}}\mathcal{L}_{\text{pred}}\big|_{\mathbf{0}} = -\frac{2}{B}\sum_{b=1}^{B} z_{\text{tgt}}^{(b)} z_{\text{ctx}}^{(b)^T} \neq \mathbf{0}$ by the coupling assumption. Hence $\mathbf{K} = \mathbf{0}$ is not a stationary point and cannot be a local minimizer; the floored penalty remains finite, $\mathcal{R}_{\text{spectral}}(\mathbf{0}) \leq \log D + \beta_{\text{norm}}(\epsilon - 1 - \log\epsilon) \approx \log D + 12.8\,\beta_{\text{norm}}$. (ii) For the unfloored barrier, $-\log r \to +\infty$ as $\|\mathbf{K}\|_F \to 0^+$ while $\mathcal{L}_{\text{pred}} \to E_{\text{tgt}} < \infty$, so $\mathcal{L}(\mathbf{K}) \to +\infty$; the identity operator satisfies $\Delta\mathcal{H}(\mathbf{I}) = \Phi_{\text{energy}}(\mathbf{I}) = 0$ with strictly finite loss, so $\mathbf{K} = \mathbf{0}$ cannot be a global minimizer in the idealization either. The divergence statement (ii) applies to the unfloored limit only; the anti-triviality guarantee for the trained objective rests on (i). □

*Proof of Proposition 2.* **(Idealized Continuous Gradient Flow Motivation: Absence of Periodic Orbits).** Suppose $\gamma(t)$ is a periodic orbit with period $T_p > 0$, i.e. $\gamma(t+T_p) = \gamma(t)$. Integrating the time derivative of $\Phi$ along the closed path yields $\int_0^{T_p} \dot{\Phi}(\gamma(t))\,dt = 0$, since $\Phi(\gamma(T_p)) = \Phi(\gamma(0))$. By the chain rule, $\dot{\Phi}(\gamma(t)) = \langle \nabla\Phi, \dot{\gamma}\rangle$, which equals $-\|\nabla\Phi(\gamma(t))\|_2^2 \leq 0$. The integral of a non-positive integrand vanishes only if $\|\nabla\Phi(\gamma(t))\|_2 \equiv 0$ on $[0, T_p]$, hence $\dot{\gamma}(t) \equiv \mathbf{0}$: $\gamma$ is an isolated fixed point, contradicting a non-constant periodic orbit. □

## S-2. Evaluation Protocol Matrix

### A. Streaming Scoring and Calibration Pseudocode

**Algorithm 1** Streaming Anomaly Scoring and Decision Calibration

**Require:** Streaming telemetry $\mathbf{X} \in \mathbb{R}^{T\times K}$, filtration $\mathcal{F}_t = \sigma(\mathbf{X}[1:t,:])$, context horizon $C = 256$, suspect horizon $S = 64$, deployment mode $\mathcal{M} \in \{\text{Causal}, \text{Buffered}\}$, calibrator $\mathcal{C}_{\text{EVT}}$ fit to nominal residuals.
**Ensure:** Point-level alarm stream $\hat{y}_t \in \{0,1\}$ for $t \in [C, T]$.

Initialize score buffer $\mathbf{s} \in \mathbb{R}^T \leftarrow \mathbf{0}$, hit counts $\mathbf{c} \in \mathbb{R}^T \leftarrow \mathbf{0}$.
**for** $t = C$ **to** $T - S$ **do**
  Extract historical context: $\mathbf{x}_{\text{ctx}} \leftarrow \mathbf{X}[t-C:t,:]$ ▷ $\mathcal{F}_t$-measurable
  Encode latent context: $z_{\text{ctx}} \leftarrow E_\theta(\mathbf{x}_{\text{ctx}})$
  Forecast future latent representation: $\hat{z}_{\text{tgt}} \leftarrow P_\psi(z_{\text{ctx}})$
  **if** $\mathcal{M} =$ Causal (0**-Lookahead Delay) then**
    Target is trailing window: $\mathbf{x}_{\text{tgt}}^{\text{causal}} \leftarrow \mathbf{X}[t-S:t,:]$ ▷ Strictly $\mathcal{F}_t$-measurable
    Encode target: $z_{\text{tgt}} \leftarrow E_{\bar{\theta}}(\mathbf{x}_{\text{tgt}}^{\text{causal}})$
    Compute trailing predictive discrepancy: $s_t \leftarrow \mathcal{D}_{\text{sal}}(\hat{z}_{\text{tgt}} - z_{\text{tgt}})$
    $\hat{y}_t \leftarrow \mathbb{I}(s_t > \tau_{\text{eff}})$ ▷ Instantaneous decision at timestamp $t$
  **else if** $\mathcal{M} =$ Buffered ($S$**-Lookahead Delay) then**
    Extract prospective suspect horizon: $\mathbf{x}_{\text{tgt}} \leftarrow \mathbf{X}[t:t+S,:]$ ▷ $\mathcal{F}_{t+S}$-measurable
    Encode target representation: $z_{\text{tgt}} \leftarrow E_{\bar{\theta}}(\mathbf{x}_{\text{tgt}})$
    Compute forward predictive discrepancy: $\mathcal{D}_t \leftarrow \mathcal{D}_{\text{sal}}(\hat{z}_{\text{tgt}} - z_{\text{tgt}})$
    **for** $\tau = t$ **to** $t + S - 1$ **do**
      Accumulate suspect overlap: $s_\tau \leftarrow s_\tau + \mathcal{D}_t$, $c_\tau \leftarrow c_\tau + 1$
    **end for**
    Finalize running average score at step $t$: $\bar{s}_t \leftarrow s_t / \max(c_t, 1)$
    $\hat{y}_t \leftarrow \mathbb{I}(\bar{s}_t > \tau_{\text{eff}})$ ▷ Finalized at timestamp $t + S$
  **end if**
**end for**
**return** Binary alarm sequence $\hat{\mathbf{y}}$

### B. Controlled Same-Backbone Comparison Table

## S-3. Auxiliary Input-Channel Saliency Gate: Specification and Studies

### A. Formal Specification

To prevent dimensional drowning in input space without target-window leakage, the strictly causal *Causal Channel Saliency Gate* computes channel dispersion strictly from historical context $\mathbf{x}_{\text{ctx}} = \mathbf{X}[t-C:t,:] \in \mathbb{R}^{C\times K}$:

$$v_k^{\text{ctx}} = \text{Var}_{\tau\in[t-C,t]}(x_{\tau,k}) = \frac{1}{C-1}\sum_{\tau=t-C}^{t}\left(x_{\tau,k} - \bar{x}_k^{\text{ctx}}\right)^2, \tag{S1}$$

$$s_k = \frac{v_k^{\text{ctx}}}{\sum_{j=1}^{K} v_j^{\text{ctx}} + \epsilon}, \quad g_k = \sigma\left(\frac{s_k - \bar{s}}{\sigma_s + \epsilon}\right), \tag{S2}$$

where $\sigma(\cdot)$ is the logistic sigmoid, $\epsilon = 10^{-6}$, $\bar{s} = 1/K$ is the uniform variance-share expectation, and $\sigma_s = \sqrt{\frac{1}{K}\sum_{k=1}^{K}(s_k - \bar{s})^2 + \epsilon}$ is the cross-channel dispersion; $(s_k - 1/K)/\sigma_s$ is the spatial z-score of channel energy. The modulation vector $\mathbf{g} \in (0,1)^K$ scales input channels symmetrically in both encoders, $\tilde{\mathbf{x}}[:,k] = \mathbf{x}[:,k]\cdot g_k$. Because $\mathbf{g}$ depends exclusively on $\mathbf{x}_{\text{ctx}}$, $\partial g_k/\partial \mathbf{x}_{\text{tgt}} \equiv \mathbf{0}$ for all $k$.

TABLE S1
MASTER EXPERIMENT MATRIX. DETAILED BREAKDOWN OF ALL EVALUATED STREAM SUBSETS, PARTITION BOUNDARIES, TEMPORAL ISOLATION GAPS, RANDOM SEEDS, AND OPERATIONAL EVALUATION REGIMES ACROSS THE EMPIRICAL INVESTIGATION.

| Experimental Suite | Streams | Domains | Partition Split | Isolation Gap | Seeds | Inference Protocol | Primary Hypotheses & Tested Endpoints |
|---|---|---|---|---|---|---|---|
| **Primary Benchmark** | 33 | 7 | 65% / 17.5% / 17.5% | 64 steps | 5/3† | Buffered ($S = 64$) | Out-of-sample held-out FPR, Point-F1, PR-AUC, event recall, and detection delay $\Delta t$ (Table II). |
| **Active Non-GHL Suite** | 19 | 6 | 65% / 17.5% / 17.5% | 64 steps | 5 | Buffered ($S = 64$) | Paired bootstrap 95% CIs and Wilcoxon signed-rank tests on active anomaly dynamics (Table II, Table I(a)). |
| **Matched Backbone** | 15 | 10 | 80% / 20% nominal | 64 steps | 1 | Buffered ($S = 64$) | Isolates observation recon vs. observation pred vs. latent pred under identical `HybridTCNEncoder` and noise injection (Table S2). |
| **Regularizer Controls** | 19 | 6 | 65% / 17.5% / 17.5% | 64 steps | 3 | Buffered ($S = 64$) | Controlled null hypothesis testing: $\lambda_{\text{ent}} = 0$, $\alpha_{\text{stress}} = 0$, and single-regime controls on fixed encoders (Table I(a)). |
| **Strictly Causal Suite** | 33 | 7 | 65% / 17.5% / 17.5% | 64 steps | 1 | Causal (0-lookahead) | Zero-lookahead streaming latency, false alarms per 1k steps ($\text{FAR}_{1k}$), and early event recall (Table S31, Table S11). |
| **EVT Multi-Parameter** | 33 | 7 | 65% / 17.5% / 17.5% | 64 steps | 1 | Buffered ($S = 64$) | Sensitivity across tail percentiles $u$, design risks $q$, kurtosis multipliers $\beta$, and non-parametric baselines (Table III, Table S13). |
| **Operational Incidents** | 3 | 3 | Continuous test | None | 1 | Standard / Sharp / Adaptive | Incident-level evaluation: Point-F1 vs. Event-F1 vs. Range-F1, and dynamic rolling SPOT drift tracking (Table S10). |

† Five seeds on the 19 non-GHL streams and GHL training loops 17–18; three seeds on the remaining twelve large GHL streams ($> 10^6$ observations each) owing to training cost.

TABLE S2
CONTROLLED SAME-BACKBONE COMPARISON ISOLATING PREDICTION SPACE UNDER AN IDENTICAL `HYBRIDTCNENCODER` ($C$=256, $S$=64, $d$=32): DATASET-MACRO MEANS ACROSS 15 STREAMS SPANNING TEN DOMAINS, UNDER CLEAN TELEMETRY AND JITTER INJECTION (SNR $\in \{\infty, 20, 10, 0$ DB$\}$).

| Model Backbone | Noise SNR | Val $\gamma_2$ | Test $\gamma_2$ | Empirical FPR | Mean False Pos. | Point-F1 | Recall |
|---|---|---|---|---|---|---|---|
| **TCN-Obs-Recon** | Clean ($\infty$) | 6.09 | 57.26 | 27.29% | 10,554.7 | 0.1977 | 0.4330 |
| | 20 dB | 6.09 | 57.23 | 27.47% | 10,587.1 | 0.1975 | 0.4337 |
| | 10 dB | 6.09 | 57.14 | 32.08% | 11,168.1 | 0.1904 | 0.4546 |
| | 0 dB | 6.09 | 57.09 | 54.10% | 17,318.2 | 0.2184 | 0.6400 |
| **TCN-Obs-Pred** | Clean ($\infty$) | 6.00 | 55.66 | 27.57% | 10,455.2 | 0.1916 | 0.4315 |
| | 20 dB | 6.00 | 56.09 | 27.93% | 10,505.6 | 0.1915 | 0.4328 |
| | 10 dB | 6.00 | 56.00 | 32.46% | 11,025.9 | 0.1984 | 0.4619 |
| | 0 dB | 6.00 | 55.66 | 50.98% | 15,558.5 | 0.2188 | 0.6414 |
| **TS-JEPA (Ours)** | Clean ($\infty$) | **0.88** | **1.77** | **11.80%** | **3,949.5** ($\downarrow 62.6\%$) | 0.0785 | 0.1460 |
| | 20 dB | **0.88** | **1.67** | **12.25%** | **4,061.3** | 0.0741 | 0.1407 |
| | 10 dB | **0.88** | **1.54** | **12.72%** | **4,248.9** | 0.0691 | 0.1300 |
| | 0 dB | **0.88** | **1.53** | **13.93%** | **4,269.0** | 0.0670 | 0.1551 |

### *B. Gating-Regime Comparison and Fault-Type Sensitivity*

Four gating regimes were trained under an identical TS-JEPA configuration ($C$=256, $S$=64, seed 42, SPOT $q$=$10^{-3}$ on held-out nominal telemetry): *context-derived gating* ($\mathbf{g} = g(\mathbf{x}_{\text{ctx}})$ applied symmetrically; strictly causal), a non-causal *target-only leaky control* ($\mathbf{g} = g(\mathbf{x}_{\text{tgt}})$, upper-bounding the benefit of violating causality), *independent* context/target gating (asymmetric; breaks latent metric alignment), and *uniform pooling* (deployed). The effect is stream-dependent in sign and magnitude: the context gate substantially improves MSL ($0.047 \rightarrow 0.227$), where anomalies concentrate in few coordinates, but degrades SMAP ($0.280 \rightarrow 0.019$), where fault signatures are distributed. Macro Point-F1 ranks uniform ($0.118$) above context gating ($0.077$), with the leaky control at $0.128$ and independent gating weakest ($0.071$). Since the leaky control adds only $+8.6\%$ relative over causal gating, most of the effect reflects quiescent-noise suppression rather than target statistics.

For per-fault-type behavior, six single-channel fault classes (flatline, stuck sensor, dropout, low-amplitude step, spike control, mid-variance flatline) were injected into six multivariate streams under four gating modes. Two findings emerge. First, the context-derived variance gate approximately doubles point recall relative to uniform pooling (macro $0.045$ vs. $0.023$) at roughly three times the nominal false-positive rate ($0.069$ vs. $0.019$); a median-absolute-deviation (MAD) variant is marginally stronger ($0.060$ recall). Second, the increase concentrates on transient signatures, while quiet faults (flatline, stuck sensor, dropout) remain weakly detected under *every* gating regime (point recall $0.02$–$0.07$). The blind spot there-

fore reflects the aggregate latent discrepancy signal on wide telemetry rather than a deficiency of gating alone.

TABLE S3
AUXILIARY INPUT-CHANNEL GATING VARIANT (NOT DEPLOYED): POINT-F1 UNDER UNIFORM, CONTEXT-GATE, TARGET-ONLY-LEAKY, AND INDEPENDENT GATING.

| Stream | Uniform (deployed) | Context Gate | Target-Only | Independent |
|---|---|---|---|---|
| NASA MSL `M-1` ($K = 55$) | 0.047 | **0.227** | 0.051 | 0.070 |
| NASA SMAP `P-3` ($K = 25$) | **0.280** | 0.019 | 0.155 | 0.055 |
| GHL Loop `17` ($K = 23$) | 0.000 | 0.000 | 0.000 | 0.000 |
| Daphnet `S01R01E1` ($K = 9$) | 0.146 | 0.061 | **0.306** | 0.158 |
| **Macro Avg** | **0.118** | 0.077 | 0.128 | 0.071 |

TABLE S4
MACRO POINT RECALL OF GATING MODES UNDER INJECTED SINGLE-CHANNEL FAULTS.

| Injected Fault Type | Uniform | Var. Gate | MAD-Gate |
|---|---|---|---|
| Flatline | 0.021 | 0.043 | **0.058** |
| Flatline (mid-var ch.) | 0.024 | 0.048 | **0.068** |
| Stuck sensor | 0.021 | 0.043 | **0.058** |
| Dropout | 0.020 | 0.043 | **0.058** |
| Low-amplitude step | 0.030 | 0.049 | **0.062** |
| $+5\sigma$ spike control | 0.023 | 0.042 | **0.059** |
| Mean nominal FPR | **0.019** | 0.069 | 0.055 |

## S-4. MATHEMATICAL DIAGNOSTICS OF PHYSICAL REGULARIZERS

To understand why the proposed regularizers enforce their intended mathematical properties yet fail to improve downstream anomaly detection, we conduct targeted dynamical diagnostic experiments.

*1) Operator-Entropy: Anti-Collapse Spectral Rank Preservation:* In Operator-Entropy JEPA, the transition operator $\mathbf{A} \in \mathbb{R}^{D\times D}$ ($D = 32$; denoted $\mathbf{K}$ in the main text, disambiguated here from input channel dimension $K$) is regularized via Von Neumann spectral entropy $S(\rho) = -\mathrm{Tr}(\rho \log \rho)$ on normalized Gram matrix $\rho = \mathbf{A}\mathbf{A}^T/\mathrm{Tr}(\mathbf{A}\mathbf{A}^T)$. Table S5 evaluates effective operator rank via the RankMe metric ($\exp(\mathcal{H})$ where $\mathcal{H}$ is the spectral entropy of singular values), representation covariance conditioning, and downstream calibrated Point-F1 across five telemetry streams.

As shown in Table S5, unconstrained Vanilla TS-JEPA and VICReg collapse across 7–8 singular modes, yielding an effective operator rank of 24.3–24.8. Operator-Entropy JEPA achieves an effective operator rank of 31.996 out of 32.0, confirming that the regularizer prevents modal collapse. However, full operator rank alone does not prevent representation collapse: the single-operator penalty acts exclusively on $\mathbf{A}$, allowing the encoder to map observations into a low-dimensional subspace ($\mathrm{Tr}(\mathbf{\Sigma}_Z) = 0.0002$) while $\mathbf{A}$ maintains full rank off-manifold.

*2) Resolution of Operator-Representation Decoupling via Dual Spectral Entropy and Lyapunov Invariance:* To resolve this decoupling, we evaluate two structural modifications: (1) dual spectral entropy coupling ($\alpha_Z$), which simultaneously penalizes non-uniformity in the representation covariance eigenspectrum $\mathbf{\Sigma}_Z$, and (2) a Discrete Lyapunov Invariance Defect ($\beta_{\text{Lyap}}$):

$$\mathcal{L}_{\text{Lyapunov}} = \frac{1}{D} \left\| \mathrm{Cov}(z_{\text{pred}}) - \mathrm{Cov}(z_{\text{tgt}}) \right\|_F^2 , \qquad \text{(S3)}$$

which penalizes rotational drift between predicted and target representation covariance structures. In addition, we evaluate the Cayley orthogonal-dissipative parameterization $W = (I + S + D)^{-1}(I - S - D)$ ($S = -S^T, D \succeq 0$), enforcing contractivity ($\|W\|_2 \leq 1.0$) by construction without soft energy penalties.

TABLE S5
EMPIRICAL SPECTRAL DIAGNOSTICS OF LEARNED TRANSITION OPERATORS AND REPRESENTATIONS ACROSS FIVE BENCHMARK TELEMETRY STREAMS ($D = 32$). OPERATOR-ENTROPY PRESERVES FULL THEORETICAL RANK ($\exp(\mathcal{H}) = 31.996$ OUT OF 32.0), PREVENTING DIMENSIONAL COLLAPSE.

| Model Architecture | Operator Rank (RankMe) | Rep. Rank | Point-F1 | Empirical FPR |
|---|---|---|---|---|
| Operator-Entropy JEPA | **31.996** | 3.564 | **0.0710** | 0.1497 |
| Representation-Entropy JEPA | 24.647 | **7.579** | 0.0479 | 0.0442 |
| Vanilla TS-JEPA (Unconstrained) | 24.807 | 3.993 | 0.0301 | **0.0371** |
| VICReg TS-JEPA | 24.339 | 5.881 | 0.0043 | 0.0273 |

TABLE S6
CONTROLLED ABLATION STUDY RESOLVING OPERATOR-REPRESENTATION DECOUPLING ACROSS EIGHT BENCHMARK TELEMETRY STREAMS UNDER STRICTLY CAUSAL TRAILING EVALUATION (SPOT $u = 0.98, q = 10^{-3}$). ENFORCING DISCRETE LYAPUNOV INVARIANCE EXPANDS REPRESENTATION RANK, CUTTING ALARM DELAY BY UP TO 73% (OPENTROPY-LYAPUNOV), WHILE DECOUPLING REPRESENTATION ENTROPY ($\alpha_Z = 0$, OPENTROPY-BOOSTED) ACHIEVES THE HIGHEST EVENT RECALL (0.4449, +33.3%). THE CAYLEY PARAMETERIZATION ACHIEVES THE HIGHEST OVERALL AUROC.

| Model Architecture | Event Recall | Delay (Steps) | Held-Out FPR | AUROC | Point-F1 |
|---|---|---|---|---|---|
| Historical OpEntropy (Single **A**-Entropy) | 0.3338 | 112.0 | 0.0241 | 0.5085 | 0.0595 |
| Vanilla TS-JEPA (Unconstrained) | 0.3763 | 85.1 | **0.0184** | 0.5039 | **0.0617** |
| OpEntropy ($\alpha_Z = 0.005$, Dual Coupling) | 0.4326 | 55.8 | 0.0328 | 0.4828 | 0.0390 |
| OpEntropy-Lyapunov ($\alpha_Z = 0.005, \beta_{\text{Lyap}} = 0.02$) | 0.4328 | **30.1** | 0.0212 | 0.5185 | 0.0501 |
| OpEntropy-Cayley-Lyapunov (Cayley $W$ + Lyap) | 0.3935 | 103.8 | 0.0336 | **0.5242** | 0.0371 |
| OpEntropy-Boosted ($\alpha_Z = 0, \beta_{\text{Lyap}} = 0.02$) | **0.4449** | 48.4 | 0.0206 | 0.4937 | 0.0498 |

As documented in Table S6, expanding representation rank via unconstrained dual spectral entropy ($\alpha_Z = 0.005$) increases nominal false alarms to 0.0328. Coupling the Discrete Lyapunov Invariance Defect aligns operator rotation with representation covariance, reducing nominal FPR by 35% (0.0212) and cutting mean event detection latency from 112.0 to 30.1 steps. Decoupling representation covariance entropy ($\alpha_Z = 0$) while maintaining the Lyapunov defect ($\beta_{\text{Lyap}} = 0.02$) and excising inert math layers (OpEntropy-Boosted) yields the highest event recall (0.4449, a +33.3% relative gain over Historical OpEntropy) at reduced nominal FPR (0.0206 vs. 0.0241), though Point-F1 (0.0498 vs. 0.0617) remains lower than unconstrained TS-JEPA. On wearable kinematics (`Daphnet:S01R01E1`), event recall reaches 0.9643 (vs. 0.5000 for TS-JEPA and 0.4643 for Historical OpEntropy). Parameterizing the transition via the Cayley transform yields the highest overall discriminative ranking (AUROC 0.5242 trailing, 0.5651 smear) and achieves complete detection (1.0000 recall) on spacecraft telemetry (`MSL:M-1`) and 0.5000 on server telemetry (`SMD:machine-1-2`) where unconstrained TS-JEPA collapses to 0.0000.

TABLE S7
CONTROLLED COVARIANCE VS. MARGINAL ANOMALY BENCHMARK. REYNOLDS-STRESS JEPA IMPROVES MARGINAL VARIANCE DETECTION OVER UNCONSTRAINED TS-JEPA (0.5838 VS. 0.3313 PR-AUC), BUT BOTH LATENT MODELS STRUGGLE ON PURE CORRELATION FLIPS.

| | Pure Covariance Shift | | Pure Marginal Shift | |
|---|---|---|---|---|
| **Model Architecture** | **PR-AUC** | **Point-F1** | **PR-AUC** | **Point-F1** |
| Reynolds-Stress JEPA | 0.0498 | 0.0000 | 0.5838 | **0.0593** |
| TS-JEPA (Unconstrained) | 0.0409 | 0.0000 | 0.3313 | 0.0000 |
| TimesNet | **0.9313** | 0.6558 | **0.9998** | 0.6267 |
| TranAD | 0.6761 | **0.7246** | 0.9343 | **0.6560** |

*3) Reynolds-Stress: Covariance Alignment vs. Marginal Perturbations:* Reynolds-Stress JEPA aligns second-order latent fluctuation statistics ($\Sigma_{z'} = \frac{1}{B}\sum z'(z')^T$) between predicted and target horizons. Table S7 compares models on synthetic telemetry isolating pure cross-channel correlation flips ($\rho : +0.85 \to -0.85$) from pure marginal variance scaling ($3.0\times$).

Table S7 indicates that on marginal variance shifts, Reynolds-Stress improves PR-AUC by $+76\%$ over TS-JEPA ($0.5838$ vs. $0.3313$). However, on pure covariance shifts without marginal changes, both latent models show low sensitivity ($0.041$–$0.050$ PR-AUC) relative to observation-space baselines ($0.676$–$0.931$). Standard temporal pooling across channels averages out instantaneous phase alignments, preventing the second-moment loss from isolating cross-channel relationship shifts unless explicit pairwise relational graphs are modeled.

*a) Covariance Estimator Ablation (9 Held-Out Streams):* To determine whether the choice of empirical covariance formulation affects the regularizer's null impact, Table S8 compares three covariance estimators across nine held-out telemetry streams: (1) the deployed instantaneous rank-1 outer product plus ridge penalty, $\mathbf{\Sigma}_{\text{obs}} = (\mathbf{P}z_{\text{tgt}})(\mathbf{P}z_{\text{tgt}})^\top + 10^{-3}\mathbf{I}$; (2) a centered temporal-window covariance over suspect steps, $\mathbf{\Sigma}_{\text{obs}} = \frac{1}{S}\sum_{s=1}^{S}(\mathbf{P}z_s - \bar{z})(\mathbf{P}z_s - \bar{z})^\top + 10^{-3}\mathbf{I}$; and (3) a batch-level empirical covariance across parallel windows. Across the nine benchmark streams under standard SPOT calibration ($u = 0.98, q = 10^{-3}$, buffered regime), macro Point-F1 scores are $0.188$ (instantaneous rank-1), $0.179$ (centered temporal-window), and $0.178$ (batch covariance). Point-F1 differences remain within $|\Delta\text{F1}| \leq 0.009$, and a paired Wilcoxon signed-rank test between instantaneous and temporal-window estimators yields $p = 0.4375$ (non-significant). The regularizer's null downstream effect is therefore invariant to the covariance estimator formulation.

*4) Potential-Flow: Conservative vs. Circulating Dynamics:* Potential-Flow JEPA constrains transition velocities to curl-free potential gradients ($\hat{v} = -\nabla\Phi$). Table S9 evaluates Potential-Flow against Helmholtz-JEPA (decomposing velocity into gradient plus solenoidal circulation: $\hat{v} = -\nabla\Phi + \text{curl}(A)$) and unconstrained TS-JEPA across two dynamical benchmarks: a conservative Duffing oscillator and a non-conservative Van der Pol limit-cycle attractor.

Table S9 confirms that pure curl-free potential flow is mathematically incapable of representing closed circulating orbits, yielding poor detection on Van der Pol ($0.3548$ Point-F1 vs. $0.7574$ for Helmholtz JEPA). Because physical telemetry often exhibits non-conservative, cyclic behavior (e.g., diurnal cycles, rotational turbines), enforcing irrotational dynamics severely penalizes normal circulating trajectories, explaining Potential-Flow's poor benchmark performance ($0.065$ dataset-macro F1 in Table II). Numerical Jacobian diagnostics show that while the raw spatial gradient field is curl-free (antisymmetric fraction $< 10^{-7}$), enforcing the spherical tangent projection introduces a $15.0\%$–$15.7\%$ antisymmetric curl component ($0.150$ on SMAP `P-3`, $0.157$ on MSL `M-1`). Tangent projection breaks the curl-free invariant without restoring sufficient circular capacity to track cyclic trajectories.

TABLE S8
REYNOLDS-STRESS COVARIANCE MODE ABLATION ACROSS NINE HELD-OUT TELEMETRY STREAMS (SPOT CALIBRATION, BUFFERED SMEAR REGIME). ACROSS INSTANTANEOUS RANK-1 OUTER PRODUCT, CENTERED TEMPORAL-WINDOW COVARIANCE, AND BATCH COVARIANCE, DOWNSTREAM POINT-F1 DIFFERENCES REMAIN WITHIN $|\Delta\text{F1}| \leq 0.009$ ($p = 0.4375$).

| | Instantaneous Rank-1 | | | Temporal-Window | | | Batch Covariance | | |
|---|---|---|---|---|---|---|---|---|---|
| **Stream** | **F1** | **FPR** | **PR-AUC** | **F1** | **FPR** | **PR-AUC** | **F1** | **FPR** | **PR-AUC** |
| Daphnet:S01R01E1 | 0.000 | 0.154 | 0.050 | 0.000 | 0.129 | 0.050 | 0.000 | 0.165 | 0.052 |
| Daphnet:S03R01E0 | 0.371 | 0.043 | 0.412 | 0.344 | 0.048 | 0.413 | 0.288 | 0.373 | 0.254 |
| Genesis:default | 0.000 | 0.006 | 0.005 | 0.000 | 0.001 | 0.005 | 0.000 | 0.024 | 0.003 |
| MSL:C-1 | 0.048 | 0.108 | 0.151 | 0.048 | 0.106 | 0.151 | 0.000 | 0.077 | 0.139 |
| MSL:M-1 | 0.000 | 0.247 | 0.424 | 0.000 | 0.252 | 0.425 | 0.000 | 0.133 | 0.408 |
| SMAP:A-7 | 0.853 | 0.150 | 0.959 | 0.853 | 0.151 | 0.959 | 0.860 | 0.121 | 0.965 |
| SMAP:D-3 | 0.321 | 0.016 | 0.860 | 0.321 | 0.016 | 0.861 | 0.323 | 0.011 | 0.849 |
| metro:traffic-volume | 0.001 | 0.019 | 0.001 | 0.001 | 0.015 | 0.001 | 0.002 | 0.016 | 0.001 |
| swan:sf | 0.098 | 0.014 | 0.392 | 0.048 | 0.005 | 0.413 | 0.126 | 0.027 | 0.349 |
| **Macro Mean** | **0.188** | 0.084 | 0.362 | **0.179** | 0.080 | 0.364 | **0.178** | 0.105 | 0.336 |

TABLE S9
DYNAMICAL SYSTEM VALIDATION COMPARING CURL-FREE POTENTIAL-FLOW ($\hat{v} = -\nabla\Phi$), HELMHOLTZ DECOMPOSITION ($\hat{v} = -\nabla\Phi + \text{CURL}(A)$), AND UNCONSTRAINED TS-JEPA ON CONSERVATIVE HAMILTONIAN VS. NON-CONSERVATIVE LIMIT-CYCLE PHASE SPACES.

| | Conservative Duffing | | Non-Conservative Van der Pol | |
|---|---|---|---|---|
| **Model Architecture** | **PR-AUC** | **Point-F1** | **PR-AUC** | **Point-F1** |
| Helmholtz JEPA ($\nabla\Phi + \text{curl}(A)$) | **0.5985** | **0.7815** | **0.5250** | **0.7574** |
| TS-JEPA (Unconstrained) | 0.5855 | 0.7583 | 0.4577 | 0.6660 |
| Potential-Flow JEPA ($-\nabla\Phi$) | 0.1701 | 0.2036 | 0.3106 | 0.3548 |

*5) Multiscale Front-End: Dyadic-Shell Construction:* The encoder front-end constructs $J{=}4$ dyadic shells on an $N_{\text{fft}} = 256$ real-FFT grid ($F = 129$ non-negative bins $\omega_m = m/N_{\text{fft}}$). Dyadic center frequencies sit at $c_j = 2^{j-J+1}$, $j \in \{0, 1, 2, 3\}$. Unnormalized profiles combine a half-Gaussian anchored at DC ($j = 0$), log-frequency octave bells $\tilde{\Psi}_j(\omega_m) = \exp(-(\log_2 \omega_m - \log_2 c_j)^2/2\sigma^2)$ with bandwidth $\sigma = 0.6$ ($j \in \{1, 2\}$), and a plateau toward Nyquist ($j = J - 1$), normalized bin-wise into an exact partition of unity:

$$\Psi_j[\omega_m] = \frac{\tilde{\Psi}_j(\omega_m)}{\sum_{l=0}^{J-1}\tilde{\Psi}_l(\omega_m)}, \qquad \text{(S4)}$$

$$\sum_{j=0}^{J-1}\Psi_j[\omega_m] \equiv 1.0.$$

The split is applied as $\mathbf{x}_j = \text{irfft}(\Psi_j \odot \text{rfft}(\mathbf{x}))$ (PoU deviation $< 1.2 \times 10^{-7}$, reconstruction error $< 5 \times 10^{-8}$). Each shell feeds a full (non-depthwise) causal 1D convolution with kernels $k \in \{15, 9, 5, 3\}$ and causal padding $p = k - 1$, recombined by a softmax router over pooled per-shell energies, a $1{\times}1$ projection, residual scaling $\gamma = 0.1$, and LayerNorm. Because shells overlap smoothly, band en-

ergies are correlated ($\sum_j \|\mathbf{x}_j\|^2$ recovers $\approx 70\%$ of signal energy) and the finest shell concentrates near Nyquist; per-band in-octave spectral mass is $\{0.50, 0.33, 0.20, \approx 1.00\}$ versus $\{0.78, 0.50, 0.55, 0.82\}$ (Haar) and $\{0.86, 0.63, 0.64, 0.87\}$ (DB4) (filterbank frequency diagnostics).

## S-5. Operational Case Studies and Computational Profile

### A. Operational Incident Evaluation: Event Metrics and Adaptation

Evaluating telemetry solely by point-level metrics can penalize slight temporal boundary offsets in continuous anomalies. In operational systems, faults represent extended continuous incidents; a detector that identifies the event with a minor phase lag is penalized on every out-of-boundary point.

To analyze operational incident detection, Table S10 evaluates three complementary metrics across three telemetry streams: Room Occupancy (stream: `default`, 11 events), NASA SMAP (stream: `P-3`, 1 event), and CalIt2 (stream: `traffic`, 21 events). We compare strict Point-F1, Event-F1 ($P_{\text{event}}, R_{\text{event}}$), and Tatbul et al. Range-Aware F1 ($F1_{\text{range}}$) across three operational inference configurations: Standard (middle-window, static EVT), Sharp (dense causal, static EVT), and Adaptive (dense, dynamic rolling SPOT).

Table S10 highlights three operational characteristics: First, static EVT thresholding achieves 100% event recall on Room Occupancy and SMAP (`P-3`), but generates over 4,000 false alarm points on Room Occupancy due to unmodeled diurnal baseline shifts, causing point precision to collapse to $20.0\%$. Second, dynamic rolling SPOT suppresses false alarms by $21\times$ (from 4,705 to 223 points) and raises precision to $29.9\%$, but event recall drops from 100% to $63.6\%$. Tracking non-stationary drift via rolling quantiles creates an operational trade-off between alarm fatigue and missed events. Third, on transient burst streams (CalIt2, 21 short spikes), models detect only $4.8\%$–$23.8\%$ of events under rolling SPOT, demonstrating that short pulses remain difficult to distinguish from ambient noise.

### B. Five-Stream Causal-vs-Buffered Operational Comparison

Table S11 compares operational latency, false-alarm rates, and detection delay across five matched telemetry streams (NASA SMAP `P-3`, SMD `machine-1-2`, NASA MSL `M-1`, Daphnet `S01R01E1`, and GECCO `water_quality`) under the strictly causal (0-lookahead) and buffered (64-lookahead) regimes defined in Section V-D.

Under strictly causal streaming, TS-JEPA maintains an empirical false-positive rate of $4.18\%$ (41.76 false alarms per 1,000 points), generating an average of 1,685.6 false alarms per stream. In contrast, TimesNet and TranAD generate 16,952.0 and 16,637.6 false alarms (empirical FPRs of $30.06\%$ and $20.81\%$). On GECCO water telemetry (113,000 test points), TimesNet and TranAD trigger over 81,500 false alarms ($99.6\%$ FPR), whereas TS-JEPA triggers zero false alarms.

Two methodological trade-offs govern these results: First, the false-alarm reduction under strict causality comes at the expense of early event capture. TS-JEPA detects $10.7\%$ of anomaly events within $\tau \leq 64$ steps under causal streaming, compared to $22.9\%$ for TimesNet and $32.9\%$ for TranAD. Second, the inherent target-window emission latency of $S-1 = 63$ steps converts to physical time according to telemetry sampling rates: for wearable gait accelerometry (Daphnet, 64 Hz), 63 steps corresponds to 0.98 s, permitting near-immediate clinical alerting; for spacecraft telemetry (NASA SMAP and MSL at 0.1 Hz) and industrial flow loops (GHL at 0.1 Hz), 63 steps corresponds to 10.5 min; for municipal water quality telemetry (GECCO, 1 sample/min), the latency is 63 min; and for municipal traffic monitoring (Metro, 1 sample/h), 63 steps represents an operational delay of 63 h. When mission constraints demand immediate zero-lookahead alarm emission, strictly causal trailing scoring must be used, accepting the sensitivity trade-off shown in Table S11.

### C. Computational Profile: Complexity, Latency, and Memory Footprint

Table S12 benchmarks parameter complexity, peak GPU memory footprint (VRAM), forward inference latency, and streaming throughput across all evaluated architectures on an NVIDIA RTX 5090 GPU (input dimension $K = 25$, batch size 32, context $C = 256$, suspect $S = 64$).

As shown in Table S12, Operator-Entropy JEPA possesses fewer trainable parameters (285,733) than unconstrained TS-JEPA (289,061), because its explicit Koopman transition matrix $\mathbf{A} \in \mathbb{R}^{32\times 32}$ (1,024 weights) replaces the multi-layer MLP predictor. All JEPA variants maintain a compact peak GPU memory footprint of 183.8–185.3 MB, consuming nearly $7\times$ less memory than TranAD (1,280.0 MB), whose self-attention maps scale quadratically with window length. Forward inference latency for the JEPA family ranges between 0.60–0.84 ms per window (1,184–1,647 windows per second). While TimesNet (0.29 ms) and TranAD (0.19 ms) achieve faster forward inference, the JEPA family operates well within sub-millisecond operational streaming deadlines on GPU hardware. These measurements characterize workstation accelerator throughput; on resource-constrained embedded processors or flight microcontrollers lacking dedicated tensor cores, sequential end-to-end latency depends directly on clock frequencies and memory bandwidth, motivating integer quantization (Section VIII) for edge deployment.

## S-6. Extended EVT Sensitivity and Failure-Mechanism Decomposition

Table S13 provides a multi-parameter sensitivity analysis of the extreme-value calibration procedure across varying initial tail thresholds $u$, theoretical design risk levels $q$, and kurtosis-adaptive coefficients $\beta$. Three findings emerge: First, varying the initial threshold from $u = 0.95$ to $u = 0.99$ produces negligible changes in out-of-sample test FPR ($17.27\%$–$17.36\%$) and Point-F1 (0.1028–0.1030), indicating that GPD tail estimates are stable with respect to initial tail percentile selection. Second, scaling the theoretical design risk $q$ by two orders of magnitude (from $10^{-2}$ to $10^{-4}$) shifts validation FPR from $0.0074$ to $0.00009$, but test FPR shifts by less than $1.7\%$ (from

TABLE S10
OPERATIONAL INCIDENT DETECTION COMPARISON ACROSS REPRESENTATIVE TELEMETRY STREAMS: ROOM OCCUPANCY (STREAM: `DEFAULT`, 11 GROUND-TRUTH EVENTS), NASA SMAP (STREAM: `P-3`, 1 THRUSTER FAULT), AND CALIT2 (STREAM: `TRAFFIC`, 21 PEDESTRIAN BURST EVENTS). WE CONTRAST STANDARD (MIDDLE, STATIC EVT), SHARP (DENSE, STATIC EVT), AND ADAPTIVE (DENSE, DYNAMIC SPOT) INFERENCE MODES.

| Evaluated Stream | Operational Configuration | Pt-F1 | Pt-Prec | Evt-F1 | Evt-Prec | Evt-Rec | Rng-F1 | False Pos. |
|---|---|---|---|---|---|---|---|---|
| *Room Occupancy: Stream `default` (Environmental Sensing, 11 Ground-Truth Incidents)* | | | | | | | | |
| `operator_entropy` | Standard (Middle, Static EVT) | 0.3728 | 22.9% | 0.5238 | 35.5% | 100% | 0.3708 | 3,842 |
| `operator_entropy` | Sharp (Dense, Static EVT) | 0.3329 | 20.0% | **0.9167** | **84.6%** | **100%** | 0.3329 | 4,705 |
| `operator_entropy` | Adaptive (Dense, Dynamic SPOT) | 0.1273 | 29.9% | 0.6364 | 63.6% | 63.6% | 0.3699 | **223** ($\downarrow 21\times$) |
| `potential_flow` | Sharp (Dense, Static EVT) | 0.3431 | 20.7% | **0.9167** | **84.6%** | **100%** | 0.3431 | 4,496 |
| `potential_flow` | Adaptive (Dense, Dynamic SPOT) | 0.0441 | 10.2% | 0.5714 | 60.0% | 54.5% | 0.1636 | **291** ($\downarrow 15\times$) |
| *NASA SMAP: Stream `P-3` (Spacecraft Telemetry, 1 Continuous Fault Segment)* | | | | | | | | |
| `operator_entropy` | Standard (Middle, Static EVT) | 0.0087 | 14.6% | 0.1818 | 10.0% | 100% | 0.2266 | 35 |
| `operator_entropy` | Sharp (Dense, Static EVT) | 0.1274 | 27.3% | 0.1538 | 8.3% | 100% | **0.3634** | 295 |
| `potential_flow` | Adaptive (Dense, Dynamic SPOT) | 0.0110 | 6.9% | **0.3333** | 20.0% | **100%** | 0.1213 | **108** |
| *CalIt2: Stream `traffic` (Pedestrian Entrance Flow, 21 Transient Event Clusters)* | | | | | | | | |
| `operator_entropy` | Standard (Middle, Static EVT) | 0.0131 | 1.9% | 0.0588 | 7.7% | 4.8% | 0.0216 | 53 |
| `operator_entropy` | Adaptive (Dense, Dynamic SPOT) | **0.1368** | 8.7% | **0.3226** | 50.0% | 23.8% | **0.1274** | 357 |
| `potential_flow` | Adaptive (Dense, Dynamic SPOT) | 0.0903 | 14.3% | 0.0909 | **100.0%** | 4.8% | 0.0714 | **42** |

TABLE S11
OPERATIONAL LATENCY AND DETECTION FIDELITY UNDER STRICTLY CAUSAL (0-LOOKAHEAD DELAY) VERSUS BUFFERED WINDOW (64-LOOKAHEAD DELAY) INFERENCE ACROSS 5 REPRESENTATIVE TELEMETRY STREAMS. UNDER CAUSAL STREAMING, TS-JEPA ACHIEVES AN EMPIRICAL FALSE-POSITIVE RATE OF 4.18% ($\text{FAR}_{1\text{K}} = 41.76$ FALSE ALARMS PER 1K POINTS), COMPARED TO 30.06% FOR TIMESNET AND 20.81% FOR TRANAD.

| Inference Setting | Model | Empirical FPR | $\text{FAR}_{1\text{k}}$ (/1k pts) | False Positives | Point-F1 | Precision | Mean Delay ($\Delta t$) | Recall ($\tau \leq 64$) |
|---|---|---|---|---|---|---|---|---|
| **Strictly Causal** (0-Lookahead) | TS-JEPA | **0.0418** | **41.76** | **1,685.6** | 0.0688 | **0.1581** | 130.3 | 0.1071 |
| | TimesNet | 0.3006 | 300.63 | 16,952.0 | 0.0774 | 0.1525 | 251.2 | 0.2286 |
| | TranAD | 0.2081 | 208.13 | 16,637.6 | **0.0881** | 0.1255 | **43.2** | **0.3286** |
| **Buffered Window** (64-Lookahead) | TS-JEPA | **0.1514** | **151.39** | **2,696.0** | 0.0431 | 0.0355 | 89.6 | 0.1357 |
| | TimesNet | 0.2800 | 280.03 | 16,916.8 | **0.0807** | 0.1404 | 235.3 | 0.2571 |
| | TranAD | 0.1998 | 199.82 | 16,339.6 | 0.0197 | **0.2284** | **32.8** | **0.2786** |

TABLE S12
COMPUTATIONAL PROFILE, MEMORY CONSUMPTION, LATENCY, AND THROUGHPUT BENCHMARKED ON AN NVIDIA RTX 5090 GPU ($K = 25$, WINDOW SIZE 320).

| Model Architecture | Parameters | VRAM (MB) | Latency (ms/win) | Infer (win/s) | Train (win/s) |
|---|---|---|---|---|---|
| TS-JEPA | 289,061 | **183.8** | 0.6072 | 1,646.8 | 612.8 |
| Operator-Entropy JEPA | **285,733** | 183.9 | 0.6939 | 1,441.0 | 553.2 |
| Reynolds-Stress JEPA | 295,953 | 184.0 | 0.7456 | 1,341.2 | 535.8 |
| Potential-Flow JEPA | 338,117 | 184.4 | 0.8445 | 1,184.2 | 400.2 |
| Helmholtz JEPA | 416,391 | 185.3 | 0.8378 | 1,193.6 | 526.6 |
| TimesNet | 648,553 | 279.6 | **0.2908** | 3,439.4 | 1,161.4 |
| TranAD | 205,746 | 1,280.0 | **0.1897** | **5,272.2** | **1,554.4** |

TABLE S13
SENSITIVITY ANALYSIS OF EXTREME VALUE THEORY (SPOT) CALIBRATION PARAMETERS ACROSS BENCHMARK STREAMS. WE EVALUATE THE SENSITIVITY OF EMPIRICAL FALSE POSITIVE RATES (ON HELD-OUT NOMINAL VALIDATION SPLITS AND TEST SEQUENCES) AND POINT-F1 ACROSS INITIAL TAIL QUANTILES $u \in \{0.95, 0.98, 0.99\}$, DESIGN RISK PARAMETERS $q \in \{10^{-2}, 10^{-3}, 10^{-4}\}$, AND KURTOSIS ADAPTATION COEFFICIENTS $\beta \in \{0.0, 0.083, 0.167, 0.25, 0.50\}$ ALONGSIDE NON-PARAMETRIC BASELINES.

| Threshold Calibration Method | Design $q$ | Tail $u$ | Val FPR | Test FPR | Point-F1 |
|---|---|---|---|---|---|
| ***Initial Tail Quantile and Risk Level Sensitivity (Static SPOT)*** | | | | | |
| Static SPOT ($u = 0.95, q = 10^{-4}$) | $10^{-4}$ | 0.95 | 0.00009 | 0.1727 | 0.1029 |
| Static SPOT ($u = 0.95, q = 10^{-3}$) | $10^{-3}$ | 0.95 | 0.00190 | 0.1762 | 0.1030 |
| Static SPOT ($u = 0.95, q = 10^{-2}$) | $10^{-2}$ | 0.95 | 0.00741 | 0.1895 | 0.1020 |
| Static SPOT ($u = 0.98, q = 10^{-4}$) | $10^{-4}$ | 0.98 | 0.00009 | 0.1733 | 0.1028 |
| Static SPOT ($u = 0.98, q = 10^{-3}$) (Baseline) | $10^{-3}$ | 0.98 | 0.00190 | 0.1764 | 0.1030 |
| Static SPOT ($u = 0.98, q = 10^{-2}$) | $10^{-2}$ | 0.98 | 0.00743 | 0.1886 | 0.1022 |
| Static SPOT ($u = 0.99, q = 10^{-4}$) | $10^{-4}$ | 0.99 | 0.00017 | 0.1736 | 0.1030 |
| Static SPOT ($u = 0.99, q = 10^{-3}$) | $10^{-3}$ | 0.99 | 0.00210 | 0.1763 | 0.1029 |
| Static SPOT ($u = 0.99, q = 10^{-2}$) | $10^{-2}$ | 0.99 | 0.00703 | 0.1862 | 0.1029 |
| ***Kurtosis-Adaptive Multiplier Sensitivity*** ($\tau_{\text{eff}} = \tau(1 + \beta \tanh \gamma_2)$) | | | | | |
| Kurtosis-Adaptive ($\beta = 0.000$, Standard) | $10^{-3}$ | 0.98 | 0.00190 | 0.1764 | 0.1030 |
| Kurtosis-Adaptive ($\beta = 0.083$) | $10^{-3}$ | 0.98 | 0.27780 | 0.2241 | 0.0973 |
| Kurtosis-Adaptive ($\beta = 0.167$) | $10^{-3}$ | 0.98 | 0.30463 | 0.2686 | 0.0964 |
| Kurtosis-Adaptive ($\beta = 0.250$) | $10^{-3}$ | 0.98 | 0.33991 | 0.3813 | 0.0995 |
| Kurtosis-Adaptive ($\beta = 0.500$) | $10^{-3}$ | 0.98 | 0.54032 | 0.6096 | 0.2560 |
| Moors Quantile Kurtosis ($\beta = 0.167$) | $10^{-3}$ | 0.98 | 0.19350 | 0.1848 | 0.1024 |
| ***Non-Parametric Validation Baselines*** | | | | | |
| Max-of-Validation | – | – | **0.00000** | **0.1774** | 0.1029 |
| Empirical 99.5th Percentile | – | – | 0.00544 | 0.1837 | 0.1028 |
| Gaussian 3-Sigma ($\mu + 3\sigma$) | – | – | 0.00882 | 0.1879 | 0.1016 |

18.95% to 17.27%), showing poor risk transfer under this protocol, without identifying drift separately from tail-sample uncertainty. Third, increasing the kurtosis adjustment coefficient $\beta$ systematically degrades performance: at $\beta = 0.083$, held-out validation FPR jumps from 0.0019 to 0.2778, and at $\beta = 0.50$, it collapses to 0.5403 (test FPR 60.96%), flooding the system with alarms. Non-parametric validation rules (such as Max-of-Validation and Empirical 99.5th Percentile) achieve comparable or superior nominal calibration (0.0000–0.0054 validation FPR) without requiring parametric tail assumptions.

To diagnose why extreme-value calibration breaks down on operational telemetry, Table S14 isolates four statistical mechanisms across benchmark telemetry streams:

1) **Residual Autocorrelation and Extremal Clustering:** The Pickands-Balkema-de Haan theorem [1] requires mutually independent tail exceedances. However, reconstructive observation residuals exhibit serial correlation, with lag-1 autocorrelation $\rho_1 = 0.983 \pm 0.014$ for TimesNet and $\rho_1 = 0.990 \pm 0.007$ for TranAD. Using the

TABLE S14
EMPIRICAL DECOMPOSITION OF EXTREME VALUE CALIBRATION FAILURE (6-STREAM PILOT DIAGNOSTIC COHORT). WE ISOLATE FOUR STATISTICAL MECHANISMS OF THRESHOLD BREAKDOWN ACROSS A PRELIMINARY 6-STREAM PILOT DIAGNOSTIC COHORT (NASA SMAP P-3, SMD MACHINE-1-2, NASA MSL M-1, DAPHNET S01R01E1, GECCO WATER_QUALITY, AND SWAN SF; DISTINCT FROM THE PRIMARY 19-STREAM HELD-OUT BENCHMARK): (1) RESIDUAL AUTOCORRELATION ($\rho_1$) AND EXTREMAL CLUSTERING ($\theta, 1/\theta$), WHICH REDUCE THE EFFECTIVE INDEPENDENT TAIL SAMPLE SIZE ($N_{\text{EFF}} = \theta N_u$); (2) NON-STATIONARY DISTRIBUTION DRIFT BETWEEN CALIBRATION AND HELD-OUT EVALUATION SPLITS (KOLMOGOROV-SMIRNOV $D_{\text{KS}}$ AND RELATIVE 98TH-PERCENTILE DRIFT $\Delta q_{98}\%$); AND (3) ACTUAL HELD-OUT NOMINAL FALSE-POSITIVE RATE AND EMPIRICAL RISK OVERSHOOT RELATIVE TO THEORETICAL DESIGN RISK $q = 10^{-3}$ (0.1%). OBSERVATION-SPACE MODELS EXHIBIT LARGE DISTRIBUTION DRIFT ($\Delta q_{98} = +470\%$) AND SEVERE CLUSTERING ($1/\theta \approx 12.7$–$13.3$ STEPS), CAUSING A 43× TO 47× OVERSHOOT OF DESIGN RISK. IN CONTRAST, LATENT PREDICTION COMPRESSES DISTRIBUTION DRIFT ($\Delta q_{98} = -0.36\%$), REDUCING RISK OVERSHOOT BY AN ORDER OF MAGNITUDE (FPR 0.33%).

| **Architecture & Objective** | **Autocorr. ($\rho_1$)** (Lag-1) | **Extremal Index ($\theta$)** (Ferro-Segers) | **Mean Cluster** ($1/\theta$ steps) | **KS Drift ($D_{\text{KS}}$)** (Cal vs. Eval) | **Held-Out FPR** (Nominal Target: 0.1%) | **Risk Overshoot** ($\text{FPR}_{\text{eval}}/10^{-3}$) |
|---|---|---|---|---|---|---|
| TimesNet (Reconstructive) | $0.983 \pm 0.014$ | $0.528 \pm 0.518$ | $12.7 \pm 17.1$ | $0.357 \pm 0.178$ | $4.32\% \pm 7.60\%$ | $43.2 \times \pm 76.0\times$ |
| TranAD (Reconstructive) | $0.990 \pm 0.007$ | $0.484 \pm 0.483$ | $13.3 \pm 17.1$ | $0.388 \pm 0.284$ | $4.76\% \pm 10.43\%$ | $47.6 \times \pm 104.3\times$ |
| TS-JEPA (Latent Predictive) | $0.820 \pm 0.135$ | $0.483 \pm 0.356$ | $5.2 \pm 5.8$ | $0.154 \pm 0.069$ | $0.33\% \pm 0.59\%$ | $3.3 \times \pm 5.9\times$ |

Ferro-Segers extremal index estimator [2], observation exceedances form persistent temporal clusters with mean cluster sizes of $1/\theta = 12.7$ to 13.3 steps (reaching 45.0 steps on GECCO water quality telemetry). Clustering collapses the effective independent tail sample size ($N_{\text{eff}} = \theta N_u$) from $N_u = 77$–91 observations down to $N_{\text{eff}} \approx 2$–6 independent clusters. In contrast, latent prediction compresses lag-1 autocorrelation to $\rho_1 = 0.820 \pm 0.135$ and mean cluster size to $5.2 \pm 5.8$ steps.

2) **Non-Stationary Distribution Drift:** Non-stationarity drives a wedge between calibration and held-out evaluation splits. Two-sample Kolmogorov-Smirnov tests reveal substantial drift in reconstructive models ($D_{\text{KS}} = 0.357 \pm 0.178$ for TimesNet; $D_{\text{KS}} = 0.388 \pm 0.284$ for TranAD; $p < 10^{-13}$ across streams). This drift shifts the 98th percentile of nominal residuals by $+470.8\%$ on TimesNet and $+48.6\%$ on TranAD between splits. Conversely, TS-JEPA maps temporal sequences into an abstract representation space where environmental shifts are compressed, yielding less than half the KS distance ($D_{\text{KS}} = 0.154 \pm 0.069$) and near-zero relative tail drift ($\Delta q_{98} = -0.36\% \pm 14.1\%$).
3) **GPD Goodness-of-Fit Degradation:** Cramér-von Mises tests for Generalized Pareto tail convergence reject the stationary GPD null hypothesis ($p < 0.05$) across autocorrelated observation streams. Because parameter estimation fits a stationary tail model to non-stationary, autocorrelated bursts, parametric SPOT thresholds provide no advantage over simple empirical validation percentiles.
4) **Empirical Risk Overshoot:** Against the specified design risk $q = 10^{-3}$ (0.1%), TimesNet generates a held-out nominal FPR of $4.32\% \pm 7.60\%$ (43.2× overshoot), and TranAD produces $4.76\% \pm 10.43\%$ (47.6× overshoot). By suppressing residual drift and tail clustering, TS-JEPA curbs the empirical nominal FPR to $0.33\% \pm 0.59\%$ (3.3× overshoot), reducing the discrepancy between theoretical risk and operational deployment by an order of magnitude.

## S-7. BENCHMARK SUITE AND STREAM DETAIL

## S-8. REVISION EXPERIMENTS

This section details the controlled revision experiments requested during review: the Experiment A controlled score-conditioning study isolating geometry from representation dynamics, operating-characteristic analyses across nominal budgets and realized FPR, controlled-drift calibration stress tests, baseline cross-validation against published benchmarks, and inference-path ablation diagnostics.

### *A. Controlled Score-Conditioning Study (Experiment A Full Contrast)*

To resolve whether low nominal false-positive rates originate from mechanical score boundedness or learned latent representation dynamics (Experiment A), Table S19 provides a complete side-by-side comparison across 18 models and controls under identical 19-stream held-out SPOT calibration ($u = 0.98, q = 10^{-3}$, buffered regime).

Two distinct operational mechanisms are evident:

1) **Bounded score geometry compresses tail kurtosis ($\gamma_2$):** Imposing LayerNorm on low-dimensional projection bottlenecks mechanically prevents extreme residual excursions. The bounded observation control `TCN-ObsPred-LN` achieves $\gamma_2 = 0.14$ and `PCA-LinPred-LN` achieves $\gamma_2 = 0.87$ (matching or exceeding `TS-JEPA`'s $\gamma_2 = 1.14$), whereas unnormalized observation forecasting (`TCN-ObsPred`) and reconstruction (`TCN-ObsRecon`) exhibit severe excess kurtosis ($\gamma_2 = 8.24$ and 8.31).
2) **Learned representation dynamics are essential for out-of-sample threshold transfer and discriminative recall:** Kurtosis compression alone does not prevent threshold breakdown across temporal blocks. Under identical SPOT calibration, `TCN-ObsPred-LN` suffers an out-of-sample nominal FPR of 0.186, matching `PCA-LinPred-LN` (0.186), compared to 0.080 for `TS-JEPA` and 0.078 for `Reynolds`. Training-free and linear controls collapse in downstream discriminative quality: `IsoForest` captures only 7.0% of events (Point-F1 0.041), and `PCA-LinPred-LN` captures 32.0% (Point-F1 0.104), compared to 47.3% Event Recall (Point-F1 0.234) for

TABLE S15
BENCHMARK SUITE SPECIFICATIONS: 45 MATCHED TELEMETRY STREAMS ACROSS 11 DATASETS (>5.06M OBSERVATIONS).

| Dataset | Physical Sensing Domain | Streams (Held / Tot) | Dim $K$ | $N_{\text{train}}$ | $N_{\text{test}}$ | Anom % ($\pi$) | Primary Anomaly Typologies | Data Partition Protocol |
|---|---|---|---|---|---|---|---|---|
| ***Domain 1: Satellite and Spacecraft Remote Telemetry*** | | | | | | | | |
| NASA SMAP [3] | Soil Moisture Satellite Subsystems | 6 / 6 | 25 | 17,022 | 50,834 | 15.18% | Subsystem power drops, thermal drifts, cmd faults | 65/17.5/17.5 held-out nominal split |
| NASA MSL [3] | Curiosity Rover Robotic Actuators | 4 / 6 | 55 | 12,331 | 13,566 | 22.88% | Actuator stalls, motor spikes, sensor freezes | 65/17.5/17.5 held-out (2 short excluded) |
| SWAN-SF [4] | Solar Photospheric Magnetograms | 1 / 1 | 38 | 19,672 | 72,000 | 27.92% | Active region magnetic field eruptive bursts | 65/17.5/17.5 held-out nominal split |
| ***Domain 2: Wearable Biometrics and Kinematics*** | | | | | | | | |
| Daphnet FoG [5] | Wearable Body Sensor Accelerometry | 6 / 6 | 9 | 48,240 | 217,344 | 13.34% | Freezing-of-Gait motor breakdown, limb tremors | 65/17.5/17.5 held-out nominal split |
| ***Domain 3: Industrial Processes and Cyber-Physical Systems*** | | | | | | | | |
| GHL Heating Loop | Thermal and Fluid Transport Loop | 14 / 14 | 16 | 1,645,137 | 1,154,014 | 1.35% | Valve sticking, heater failures, pressure surge | 65/17.5/17.5 held-out nominal split |
| Genesis Automation | Robotic Pick-and-Place Demonstrator | 1 / 1 | 18 | 3,244 | 9,332 | 0.30% | Mechanical manipulation jams, gripper slip | 65/17.5/17.5 held-out nominal split |
| GECCO Water IoT | Hydraulic and Water Distribution | 0 / 1 | 9 | 27,560 | 83,113 | 1.54% | Chemical contamination, line bursts, pH drops | Operational case study ($> 10^6$ steps) |
| ***Domain 4: Smart Infrastructure and Environmental Sensing*** | | | | | | | | |
| Room-Occupancy [6] | Ambient Environmental Sensing | 0 / 2 | 5 | 2,565 | 14,317 | 19.31% | Room-occupancy transitions, airflow anomalies | Incident case study (nominal too short) |
| CICIDS Edge Network | Edge Gateway Flow Telemetry | 0 / 6 | 72 | 278,722 | 1,343,920 | 23.91% | DDoS volume surges, brute force, port scans | Extended archive ($> 10^6$ steps) |
| Callt2 [7] | Optical People-Flow Counter | 0 / 1 | 2 | 940 | 4,032 | 3.42% | Crowd attendance surges, building ingress bursts | Incident case study (nominal too short) |
| Metro Traffic | Highway Loop Detector Volume | 1 / 1 | 5 | 24,072 | 24,102 | 0.13% | Highway congestion, accidents, severe slowdowns | 65/17.5/17.5 held-out nominal split |
| **Total / Benchmark Suite** | **4 Diverse Physical Domains** | **33 / 45** | **2–72** | **2,079,507** | **2,986,574** | **13.72% (avg)** | **Structural, contextual, and point faults** | **33 Primary Held-Out / 12 Case Studies** |

TABLE S16
PER-DATASET PRECISION-RECALL AREA UNDER THE CURVE (PR-AUC / AVERAGE PRECISION) ACROSS ALL 7 EVALUATED TELEMETRY BENCHMARKS (33 MATCHED CHANNELS). THE BEST RESULT FOR EACH DATASET IS IN **BOLD**, AND THE SECOND BEST IS UNDERLINED. CONTINUOUS RANKING EVALUATES THE UNTHRESHOLDED PREDICTIVE SCORES.

| Dataset | Physical Telemetry Domain | Streams | TS-JEPA (Control) | Operator-Entropy | Reynolds-Stress | Potential-Flow | TimesNet | TranAD |
|---|---|---|---|---|---|---|---|---|
| Daphnet | Wearable Parkinsonian Kinematics | 6 | 0.137 | 0.150 | 0.151 | 0.144 | **0.343** | 0.280 |
| GHL | Gasoil Heating Loop Dynamics | 14 | **0.016** | 0.015 | 0.011 | 0.013 | 0.009 | 0.010 |
| Genesis | Pick-and-Place Automation | 1 | **0.053** | 0.009 | 0.005 | 0.005 | 0.020 | 0.006 |
| MSL | NASA Mars Science Laboratory | 4 | 0.427 | **0.445** | 0.413 | 0.281 | 0.310 | 0.294 |
| SMAP | NASA Soil Moisture Active Passive | 6 | 0.539 | **0.591** | 0.522 | 0.261 | 0.463 | 0.350 |
| Metro | Interstate Highway Traffic (I-94) | 1 | 0.001 | 0.001 | 0.001 | 0.001 | 0.002 | **0.002** |
| SWAN | Solar Flare Magnetic Telemetry | 1 | 0.519 | 0.370 | 0.397 | 0.325 | 0.292 | **0.562** |
| **Dataset-Macro Mean** | Across All 7 Benchmarks | 33 | **0.242** | 0.227 | 0.214 | 0.147 | 0.206 | 0.216 |

TABLE S17
PER-DATASET UNADJUSTED POINT-F1 ACROSS ALL 7 EVALUATED TELEMETRY BENCHMARKS (33 MATCHED STREAMS) UNDER HELD-OUT CALIBRATION (SPOT, $q = 10^{-3}$; UP TO FIVE SEEDS ON NON-GHL STREAMS). BEST RESULT FOR EACH DATASET IS IN **BOLD**, AND SECOND BEST IS UNDERLINED. THE BOTTOM ROW REPORTS UNWEIGHTED DATASET-MACRO AVERAGES MATCHING TABLE II.

| Dataset | Physical Sensing Domain | Streams | TS-JEPA (Control) | Operator-Entropy | Reynolds-Stress | Potential-Flow | TimesNet | TranAD |
|---|---|---|---|---|---|---|---|---|
| Daphnet | Wearable Parkinsonian Kinematics | 6 | 0.125 | 0.136 | 0.111 | 0.057 | **0.229** | 0.144 |
| GHL | Gasoil Heating Loop Dynamics | 14 | 0.000 | 0.001 | 0.001 | 0.000 | **0.022** | 0.021 |
| Genesis | Pick-and-Place Automation | 1 | **0.017** | 0.007 | 0.000 | 0.002 | 0.000 | 0.000 |
| MSL | NASA Mars Science Laboratory | 4 | 0.237 | **0.303** | 0.219 | 0.122 | 0.158 | 0.151 |
| SMAP | NASA Soil Moisture Active Passive | 6 | **0.415** | 0.401 | 0.376 | 0.117 | 0.289 | 0.167 |
| Metro | Interstate Highway Traffic (I-94) | 1 | 0.000 | 0.000 | 0.001 | **0.004** | 0.001 | 0.000 |
| SWAN | Solar Flare Magnetic Telemetry | 1 | **0.241** | 0.060 | 0.148 | 0.146 | 0.000 | 0.000 |
| **Dataset-Macro Mean** | Across All 7 Benchmarks | 33 | **0.148** | 0.130 | 0.122 | 0.064 | 0.100 | 0.069 |

TABLE S18
NATIVE-WINDOW BASELINE EVALUATION ACROSS FIVE DIVERSE BENCHMARK STREAMS (NASA SMAP: `P-3`, SMD: `MACHINE-1-2`, NASA MSL: `M-1`, DAPHNET: `S01R01E1`, GECCO: `WATER_QUALITY`). SHORTENING THE CONTEXT WINDOW TO NATIVE SIZES ($W = 10$ FOR TRANAD, $W = 96$ FOR TIMESNET) EXACERBATES RESIDUAL TAIL HEAVINESS ($\gamma_2 = 448.25$ AND $98.14$) RATHER THAN CURING IT.

| Model Configuration | Window $W$ | Test $\gamma_2$ | Mean False Pos. | Empirical FPR |
|---|---|---|---|---|
| **TimesNet (Native)** | 96 | 98.14 | 26,526.8 | 44.09% |
| **TimesNet (Long)** | 320 | 32.04 | 27,041.0 | 51.26% |
| **TranAD (Native)** | 10 | 448.25 | 18,765.4 | 24.51% |
| **TranAD (Long)** | 320 | 29.93 | 16,671.0 | 31.76% |
| **TS-JEPA (Ours)** | 320 | **3.14** | **10,678.8** | **22.57%** |

`TS-JEPA`. Nonlinear latent representations are therefore strictly required to preserve threshold transfer and discriminative fault detection.

### B. Controlled-Drift Calibration Stress Test

Table S22 evaluates calibrator reliability under nine controlled distribution-shift conditions injected into held-out nominal score sequences. The experiment is strictly risk-matched at design risk $q = 10^{-3}$ ($p_{99.9}$), not $0.5\%$: the static and rolling empirical quantiles evaluate the empirical $1-q = 0.999$ calibration quantile to match SPOT's nominal budget. While rolling calibrators control nominal FPR to $1.1\%$–$7.0\%$ under discrete step changes, variance inflations, and autocorrelation shifts, both static and rolling methods fail under continuous ramps ($0.326$–$0.385$ FPR), confirming that temporal score drift within blocks dominates tail calibration failure.

### C. Shared Intersecting Event Detection Delay Analysis

Apparent detection delay differences in unconditioned benchmarks often stem from differing event detection subsets rather than intrinsic response speed: detectors with higher false-positive rates trigger on earlier margins of diffuse events, or capture distinct anomaly subsets. To separate detector latency from event-selection bias, Table S25 evaluates detection delay on the intersection of anomaly events detected by both TS-JEPA and each evaluated model across paired stream–seed runs at $q^*$=0.05.

TABLE S19
CONTROLLED SCORE-CONDITIONING STUDY (EXPERIMENT A): RESIDUAL-DISTRIBUTION DIAGNOSTICS AND DOWNSTREAM DETECTION METRICS ON HELD-OUT NOMINAL TELEMETRY (BUFFERED REGIME, STREAM-AVERAGED OVER 19 NON-GHL STREAMS AT FIVE SEEDS). $\gamma_2$: EXCESS KURTOSIS; $q_{99.9}/q_{99}$: TAIL RATIO; $\theta$: EXTREMAL INDEX (RUNS DECLUSTERING); $N_{\text{eff}}$: EFFECTIVE INDEPENDENT EXCEEDANCES; FPR: OUT-OF-SAMPLE EXCEEDANCE RATE AT STATIC SPOT THRESHOLD ($u = 0.98, q = 10^{-3}$); POINT-F1: UNADJUSTED POINT F1; EVENT-REC: EVENT RECALL; $\Delta t$: MEAN DETECTION DELAY (STEPS) OVER DETECTED EVENTS; PR-AUC: PRECISION-RECALL AREA UNDER CURVE.

| Model Architecture | $\gamma_2$ | $q_{99.9}/q_{99}$ | $\text{ACF}_1$ | $\theta$ | $N_{\text{eff}}$ | FPR | Point-F1 | Event-Rec | Delay ($\Delta t$) | PR-AUC |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Latent-Space Predictive JEPAs (Learned Nonlinear Dynamics)*** | | | | | | | | | | |
| TS-JEPA (Unconstrained Control) | 1.14 | 2.065 | 0.991 | 0.178 | 1.7 | 0.080 | 0.234 | 0.473 | 75.9 | 0.323 |
| TS-JEPA-NoGate | 1.05 | 1.738 | 0.992 | 0.175 | 1.6 | 0.080 | 0.235 | 0.499 | 154.4 | 0.324 |
| OpEntropy | 1.39 | 2.241 | 0.989 | 0.174 | 1.7 | 0.083 | 0.238 | 0.506 | 197.4 | 0.322 |
| Reynolds | 1.28 | 2.039 | 0.989 | 0.174 | 1.8 | 0.078 | 0.204 | 0.440 | 141.0 | 0.305 |
| PotentialFlow | 1.58 | 2.379 | 0.985 | 0.187 | 1.9 | 0.093 | 0.134 | 0.422 | 95.2 | 0.236 |
| JEPA-CtxRecon (No Foresight) | 1.01 | 2.300 | 0.991 | 0.188 | 1.7 | 0.108 | 0.208 | 0.513 | 97.9 | 0.277 |
| JEPA-NoEMA (Hard-Copy Target) | 1.00 | 2.187 | 0.988 | 0.192 | 1.8 | 0.113 | 0.234 | 0.508 | 134.0 | 0.314 |
| ***Observation-Space Forecasters and Reconstructors (Unnormalized vs. Bounded Bottleneck Controls)*** | | | | | | | | | | |
| TCN-ObsPred (Unnormalized) | 8.24 | 1.209 | 0.998 | 0.121 | 1.0 | 0.310 | 0.144 | 0.555 | 214.1 | 0.310 |
| TCN-ObsPred-LN (LayerNorm 32d) | 0.14 | 1.842 | 0.992 | 0.176 | 1.6 | 0.186 | 0.131 | 0.536 | 235.5 | 0.315 |
| TCN-ObsRecon (Unnormalized) | 8.31 | 1.316 | 0.998 | 0.126 | 1.1 | 0.303 | 0.135 | 0.550 | 220.0 | 0.301 |
| TimesNet (ICLR 2023) | 8.24 | 1.309 | 0.998 | 0.120 | 1.0 | 0.299 | 0.194 | 0.578 | 52.2 | 0.329 |
| TranAD (VLDB 2022) | 10.40 | 1.220 | 0.998 | 0.122 | 1.1 | 0.307 | 0.130 | 0.490 | 234.4 | 0.287 |
| ***Linear, Nonparametric, and Median-Based Observation Controls*** | | | | | | | | | | |
| PCA-LinPred-LN (Linear Ridge) | 0.87 | 2.555 | 0.990 | 0.207 | 1.8 | 0.186 | 0.104 | 0.320 | 137.5 | 0.244 |
| PCA-Recon (Subspace) | 8.59 | 1.231 | 0.998 | 0.119 | 1.0 | 0.312 | 0.198 | 0.584 | 46.2 | 0.331 |
| Obs-MAD (Median Norm) | 1.12 | 1.428 | 0.998 | 0.127 | 1.1 | 0.316 | 0.142 | 0.552 | 117.6 | 0.290 |
| IsoForest (Isolation Trees) | -0.34 | 1.383 | 0.996 | 0.164 | 1.4 | 0.056 | 0.041 | 0.070 | 171.0 | 0.222 |
| KNN-Dist ($k$NN Distance) | 3.37 | 1.376 | 0.998 | 0.124 | 1.1 | 0.301 | 0.161 | 0.546 | 63.5 | 0.326 |
| ZScoreMax (Per-Channel Max-z) | 9.97 | 1.617 | 0.997 | 0.131 | 1.2 | 0.339 | 0.209 | 0.609 | 67.5 | 0.374 |

TABLE S20
OPERATING-CHARACTERISTIC SUMMARY ON 19 NON-GHL STREAMS (SEEDS AVERAGED WITHIN STREAMS). PARTIAL AUC = MEAN EVENT RECALL OVER DESIGN BUDGETS $\leq 0.10$; $q^*$=0.05 COLUMNS GIVE EVENT RECALL AND *realized* TEST FPR AT THAT DESIGN BUDGET.

| Model | pAUC (trail.) | pAUC (smear) | $\text{ER}_{0.05}$ (trail.) | $\text{FPR}_{0.05}$ (trail.) | $\text{ER}_{0.05}$ (smear) | $\text{FPR}_{0.05}$ (smear) |
|---|---|---|---|---|---|---|
| PotentialFlow | 0.749 | 0.528 | 0.784 | 0.108 | 0.537 | 0.273 |
| JEPA-CtxRecon | 0.748 | 0.604 | 0.779 | 0.235 | 0.610 | 0.353 |
| OpEntropy | 0.746 | 0.612 | 0.771 | 0.193 | 0.627 | 0.315 |
| TS-JEPA | 0.742 | 0.553 | 0.774 | 0.180 | 0.560 | 0.259 |
| JEPA-NoEMA | 0.741 | 0.609 | 0.765 | 0.239 | 0.620 | 0.362 |
| TS-JEPA-NoGate | 0.726 | 0.549 | 0.768 | 0.175 | 0.548 | 0.269 |
| Reynolds | 0.689 | 0.534 | 0.709 | 0.181 | 0.543 | 0.269 |
| PCA-LinPred-LN | 0.661 | 0.433 | 0.665 | 0.140 | 0.425 | 0.227 |
| PCA-Recon | 0.627 | 0.621 | 0.623 | 0.263 | 0.597 | 0.285 |
| TCN-ObsPred | 0.626 | 0.618 | 0.629 | 0.299 | 0.595 | 0.323 |
| IsoForest | 0.622 | 0.483 | 0.631 | 0.196 | 0.482 | 0.237 |
| TimesNet | 0.618 | 0.613 | 0.618 | 0.262 | 0.612 | 0.287 |
| TCN-ObsRecon | 0.618 | 0.609 | 0.620 | 0.299 | 0.590 | 0.322 |
| KNN-Dist | 0.616 | 0.596 | 0.611 | 0.265 | 0.590 | 0.317 |
| Obs-MAD | 0.611 | 0.612 | 0.604 | 0.201 | 0.606 | 0.235 |
| TranAD | 0.603 | 0.599 | 0.602 | 0.270 | 0.596 | 0.295 |
| ZScoreMax | 0.594 | 0.621 | 0.595 | 0.258 | 0.609 | 0.320 |

Across the 244 shared anomaly events detected by both models across 65 evaluation runs, TS-JEPA achieves a mean detection delay of 59.21 steps, matching TimesNet (56.91 steps) within 2.3 steps (2.30 steps difference). This contrasts with the 31.8-step gap observed under unconditioned event sets (Table II). Against TCN-ObsPred (249 intersecting events across 65 runs), the delay gap is 9.01 steps (60.35 vs. 51.33 steps). When conditioned on identical detected anomalies, the latency disparity between reconstructive observation models and latent representation forecasters largely disappears.

### D. Multi-Dataset Baseline Benchmark Analysis

Table S28 evaluates TS-JEPA against published deep anomaly detection baselines across all 11 shared benchmark datasets (130 continuous telemetry streams) using the unified out-of-sample calibration protocol (standard SPOT, $u = 0.98, q = 10^{-3}$). Two empirical findings stand out:

*Contrastive sensitivity under tail calibration*: NCAD [8], which trains against synthetic point and contextual anomalies, achieves respectable Point-Adjusted F1 (0.252 macro, 0.298 stream-level) but collapses to 0.058 macro Point-F1 (0.088 stream-level). Point adjustment masks rampant false alarms because an isolated trigger within a long anomaly awards full segment credit. When evaluated strictly point-by-point, synthetic anomalies fail to transfer to physical process deviations (yielding near-zero detection on GHL, Genesis, and Metro). Similarly, dual-attention discrepancy contrasting in DCdetector [9] and Anomaly Transformer [10] collapses to 0.070 and 0.091 macro Point-F1 due to severe tail-threshold overshoots on non-stationary baselines.

*Forecasting versus latent representation*: Observation-space forecasters (TimesNet [11]: 0.129 macro Point-F1; TranAD [12]: 0.156) outperform contrastive baselines on point metrics but remain constrained by raw observation noise. TS-JEPA achieves leading macro Point-F1 (0.160) and stream-level Point-F1 (0.186), alongside leading Point-Adjusted F1 (0.310 macro, 0.401 stream-level), indicating that predicting representations in latent space preserves fault dynamics across disparate physical domains without synthetic anomaly synthesis.

### E. Master Headline False-Positive Rate Reconciliation

To eliminate ambiguity across reported False-Positive Rates in the manuscript and supplementary material, Table S29 provides a complete reconciliation mapping every headline FPR statistic to its evaluation cohort, random seed coverage,

operational attribution regime, calibrator, aggregation level, and primary manuscript context.

### *F. Complete 17-Model Common Support Matrix on Realized FPR*

To examine detection quality without reliance on nominal calibration transfer, Table S30 provides the complete 17-model common support matrix on realized test false-positive rates $[0.10, 0.50]$ under the strictly causal trailing regime. For each model contrasted against TS-JEPA, common support requires observed test false-positive rates to cover $[0.10, 0.50]$ without extrapolation. Out of 19 non-GHL benchmark streams, models with conservative or aggressive threshold transfer support between 4 and 14 streams, discarding 5 to 15 streams.

Across all 17 paired comparisons, TS-JEPA demonstrates no statistically significant advantage over alternative models on common support: stream-level paired Wilcoxon signed-rank test $p$-values range from $0.219$ to $1.000$, and every single contrast yields a Benjamini–Hochberg adjusted $p_{\mathrm{BH}} = 1.0$. For PCA-LinPred-LN ($+0.121$ mean $\Delta\mathrm{pAUC}$), the stream-level paired Wilcoxon signed-rank test yields $p = 0.312$ with a 95% bootstrap CI spanning zero ($[-0.008, +0.271]$ across 11 supported streams). Reconstructive TimesNet yields a $+0.061$ difference across 4 supported streams ($p = 0.625$, CI $[-0.153, +0.329]$), while bounded TCN-ObsPred-LN matches TS-JEPA closely ($-0.018$, $p = 0.219$, CI $[-0.044, +0.002]$ across 9 supported streams). Common-support interpolation confirms that nominal-budget advantages do not translate into statistically significant detection gains at matched realized false-positive rates.

TABLE S21
DETECTION QUALITY AT MATCHED HELD-OUT FALSE-POSITIVE-RATE BUDGETS (STREAM-AVERAGED, $q^*$ = EMPIRICAL CALIBRATION QUANTILE). ER: EVENT RECALL; P: POINT PRECISION; $\Delta t$: MEAN DETECTION DELAY; FA/1K: FALSE ALARMS PER 1,000 NOMINAL POINTS; $\mathrm{ER}_{\mathrm{pre}}$: RECALL CREDITING PRE-ONSET ALARMS.

| Regime ($q^*$) | Model | ER | P | $\Delta t$ | FA/1k | $\mathrm{ER}_{\mathrm{pre}}$ |
|---|---|---|---|---|---|---|
| Buffered 0.001 | JEPA-CtxRecon | 0.526 | 0.211 | 104.5 | 291.3 | 0.543 |
| Buffered 0.001 | JEPA-NoEMA | 0.526 | 0.273 | 159.4 | 308.4 | 0.552 |
| Buffered 0.001 | OpEntropy | 0.523 | 0.264 | 197.3 | 247.9 | 0.529 |
| Buffered 0.001 | PCA-Recon | 0.584 | 0.208 | 45.2 | 254.4 | 0.596 |
| Buffered 0.001 | PotentialFlow | 0.411 | 0.163 | 109.5 | 206.2 | 0.452 |
| Buffered 0.001 | Reynolds | 0.442 | 0.243 | 139.6 | 213.4 | 0.456 |
| Buffered 0.001 | TCN-ObsPred | 0.582 | 0.165 | 213.3 | 289.8 | 0.593 |
| Buffered 0.001 | TCN-ObsRecon | 0.575 | 0.164 | 216.2 | 288.0 | 0.579 |
| Buffered 0.001 | TS-JEPA | 0.486 | 0.277 | 86.0 | 207.9 | 0.510 |
| Buffered 0.001 | TimesNet | 0.583 | 0.194 | 51.5 | 259.2 | 0.585 |
| Buffered 0.001 | TranAD | 0.516 | 0.130 | 128.0 | 262.6 | 0.554 |
| Buffered 0.001 | ZScoreMax | 0.609 | 0.290 | 77.8 | 283.2 | 0.662 |
| Buffered 0.005 | JEPA-CtxRecon | 0.543 | 0.195 | 102.7 | 303.6 | 0.571 |
| Buffered 0.005 | JEPA-NoEMA | 0.552 | 0.272 | 148.1 | 318.1 | 0.578 |
| Buffered 0.005 | OpEntropy | 0.536 | 0.267 | 191.7 | 259.2 | 0.544 |
| Buffered 0.005 | PCA-Recon | 0.584 | 0.209 | 44.8 | 255.9 | 0.596 |
| Buffered 0.005 | PotentialFlow | 0.452 | 0.174 | 111.3 | 218.3 | 0.487 |
| Buffered 0.005 | Reynolds | 0.476 | 0.246 | 127.0 | 220.5 | 0.492 |
| Buffered 0.005 | TCN-ObsPred | 0.584 | 0.166 | 214.6 | 291.2 | 0.596 |
| Buffered 0.005 | TCN-ObsRecon | 0.575 | 0.164 | 214.5 | 289.6 | 0.583 |
| Buffered 0.005 | TS-JEPA | 0.503 | 0.269 | 84.5 | 216.3 | 0.536 |
| Buffered 0.005 | TimesNet | 0.583 | 0.194 | 51.4 | 260.7 | 0.585 |
| Buffered 0.005 | TranAD | 0.516 | 0.130 | 127.0 | 264.6 | 0.555 |
| Buffered 0.005 | ZScoreMax | 0.609 | 0.290 | 77.6 | 286.8 | 0.662 |
| Buffered 0.010 | JEPA-CtxRecon | 0.560 | 0.192 | 98.3 | 313.5 | 0.576 |
| Buffered 0.010 | JEPA-NoEMA | 0.554 | 0.263 | 145.0 | 325.5 | 0.581 |
| Buffered 0.010 | OpEntropy | 0.540 | 0.264 | 187.6 | 265.9 | 0.550 |
| Buffered 0.010 | PCA-Recon | 0.584 | 0.210 | 44.0 | 258.1 | 0.598 |
| Buffered 0.010 | PotentialFlow | 0.468 | 0.170 | 102.9 | 227.4 | 0.503 |
| Buffered 0.010 | Reynolds | 0.488 | 0.243 | 137.1 | 227.0 | 0.504 |
| Buffered 0.010 | TCN-ObsPred | 0.584 | 0.167 | 212.5 | 293.6 | 0.598 |
| Buffered 0.010 | TCN-ObsRecon | 0.575 | 0.164 | 211.5 | 292.1 | 0.590 |
| Buffered 0.010 | TS-JEPA | 0.510 | 0.261 | 79.2 | 222.3 | 0.553 |
| Buffered 0.010 | TimesNet | 0.590 | 0.194 | 50.6 | 262.7 | 0.593 |
| Buffered 0.010 | TranAD | 0.522 | 0.131 | 124.4 | 267.4 | 0.561 |
| Buffered 0.010 | ZScoreMax | 0.609 | 0.290 | 67.4 | 295.1 | 0.662 |
| Buffered 0.050 | JEPA-CtxRecon | 0.610 | 0.202 | 81.6 | 352.7 | 0.631 |
| Buffered 0.050 | JEPA-NoEMA | 0.620 | 0.243 | 137.0 | 362.3 | 0.645 |
| Buffered 0.050 | OpEntropy | 0.627 | 0.248 | 70.7 | 314.9 | 0.640 |
| Buffered 0.050 | PCA-Recon | 0.597 | 0.235 | 51.4 | 284.5 | 0.635 |
| Buffered 0.050 | PotentialFlow | 0.537 | 0.169 | 77.0 | 272.5 | 0.584 |
| Buffered 0.050 | Reynolds | 0.543 | 0.231 | 111.2 | 269.3 | 0.572 |
| Buffered 0.050 | TCN-ObsPred | 0.595 | 0.192 | 205.3 | 323.1 | 0.637 |
| Buffered 0.050 | TCN-ObsRecon | 0.590 | 0.186 | 203.0 | 321.7 | 0.628 |
| Buffered 0.050 | TS-JEPA | 0.560 | 0.246 | 61.2 | 258.6 | 0.612 |
| Buffered 0.050 | TimesNet | 0.612 | 0.196 | 50.0 | 287.3 | 0.615 |
| Buffered 0.050 | TranAD | 0.596 | 0.145 | 146.3 | 295.3 | 0.622 |
| Buffered 0.050 | ZScoreMax | 0.609 | 0.286 | 63.5 | 319.5 | 0.662 |
| Causal 0.001 | JEPA-CtxRecon | 0.542 | 0.207 | 122.1 | 128.4 | – |
| Causal 0.001 | JEPA-NoEMA | 0.512 | 0.256 | 164.0 | 126.6 | – |
| Causal 0.001 | OpEntropy | 0.557 | 0.232 | 222.9 | 103.1 | – |
| Causal 0.001 | PCA-Recon | 0.589 | 0.217 | 206.5 | 210.2 | – |
| Causal 0.001 | PotentialFlow | 0.482 | 0.186 | 184.6 | 35.1 | – |
| Causal 0.001 | Reynolds | 0.486 | 0.249 | 159.5 | 93.0 | – |
| Causal 0.001 | TCN-ObsPred | 0.546 | 0.178 | 46.9 | 248.7 | – |
| Causal 0.001 | TCN-ObsRecon | 0.528 | 0.174 | 46.9 | 244.5 | – |
| Causal 0.001 | TS-JEPA | 0.499 | 0.261 | 123.3 | 70.0 | – |
| Causal 0.001 | TimesNet | 0.590 | 0.219 | 60.1 | 222.4 | – |
| Causal 0.001 | TranAD | 0.533 | 0.156 | 127.1 | 209.6 | – |
| Causal 0.001 | ZScoreMax | 0.587 | 0.294 | 62.4 | 237.9 | – |
| Causal 0.005 | JEPA-CtxRecon | 0.600 | 0.211 | 93.1 | 149.0 | – |
| Causal 0.005 | JEPA-NoEMA | 0.599 | 0.253 | 124.2 | 150.1 | – |
| Causal 0.005 | OpEntropy | 0.634 | 0.232 | 81.2 | 116.7 | – |
| Causal 0.005 | PCA-Recon | 0.595 | 0.216 | 203.9 | 215.3 | – |
| Causal 0.005 | PotentialFlow | 0.593 | 0.194 | 102.9 | 46.7 | – |
| Causal 0.005 | Reynolds | 0.544 | 0.243 | 120.8 | 110.4 | – |
| Causal 0.005 | TCN-ObsPred | 0.548 | 0.178 | 44.8 | 251.3 | – |
| Causal 0.005 | TCN-ObsRecon | 0.555 | 0.174 | 43.9 | 248.0 | – |
| Causal 0.005 | TS-JEPA | 0.628 | 0.275 | 104.1 | 89.2 | – |
| Causal 0.005 | TimesNet | 0.590 | 0.219 | 59.7 | 226.4 | – |
| Causal 0.005 | TranAD | 0.536 | 0.155 | 126.6 | 213.0 | – |
| Causal 0.005 | ZScoreMax | 0.587 | 0.294 | 62.4 | 237.9 | – |
| Causal 0.010 | JEPA-CtxRecon | 0.640 | 0.215 | 75.4 | 165.7 | – |
| Causal 0.010 | JEPA-NoEMA | 0.662 | 0.241 | 74.9 | 168.2 | – |
| Causal 0.010 | OpEntropy | 0.652 | 0.231 | 66.8 | 127.6 | – |
| Causal 0.010 | PCA-Recon | 0.595 | 0.219 | 45.0 | 221.3 | – |
| Causal 0.010 | PotentialFlow | 0.637 | 0.193 | 70.9 | 54.8 | – |
| Causal 0.010 | Reynolds | 0.626 | 0.250 | 97.4 | 125.2 | – |
| Causal 0.010 | TCN-ObsPred | 0.605 | 0.180 | 41.5 | 255.3 | – |
| Causal 0.010 | TCN-ObsRecon | 0.595 | 0.174 | 40.4 | 252.4 | – |
| Causal 0.010 | TS-JEPA | 0.662 | 0.264 | 85.7 | 110.3 | – |
| Causal 0.010 | TimesNet | 0.595 | 0.220 | 63.4 | 228.8 | – |

TABLE S22
CONTROLLED-DRIFT CALIBRATION STUDY: ACHIEVED HELD-OUT FPR (DESIGN $q = 10^{-3}$) UNDER INJECTED SHIFTS IN REAL RESIDUAL STREAMS (STREAM-AVERAGED OVER 16 STREAMS WITH $\geq 200$ CALIBRATION/EVALUATION WINDOWS; THE THREE SHORT MSL STREAMS ARE EXCLUDED). ROLLING VARIANTS RECALIBRATE PER EVALUATION BLOCK.

| Condition | SPOT | SPOT+decl. | Empirical ($q = 10^{-3}$) | Roll. Emp. ($q = 10^{-3}$) | Roll. SPOT |
|---|---|---|---|---|---|
| acf 0.6 | 0.079 | 0.182 | 0.188 | 0.018 | 0.051 |
| acf 0.9 | 0.113 | 0.218 | 0.232 | 0.024 | 0.055 |
| mean ramp | 0.075 | 0.182 | 0.186 | 0.326 | 0.385 |
| mean shift +0.5s | 0.075 | 0.181 | 0.186 | 0.011 | 0.040 |
| mean shift +1.0s | 0.087 | 0.209 | 0.216 | 0.011 | 0.043 |
| regime change +1s | 0.073 | 0.182 | 0.187 | 0.015 | 0.070 |
| stationary | 0.061 | 0.159 | 0.162 | 0.016 | 0.047 |
| var scale 1.5 | 0.072 | 0.172 | 0.178 | 0.016 | 0.047 |
| var scale 2.0 | 0.082 | 0.186 | 0.195 | 0.016 | 0.047 |

TABLE S23
TRAINING-FREE NONPARAMETRIC REFERENCES UNDER THE IDENTICAL HELD-OUT CALIBRATION PROTOCOL (19 NON-GHL STREAMS, FIVE SEEDS, BUFFERED REGIME).

| Model | Held-out FPR | Point-F1 | PR-AUC | Event Recall | $\Delta t$ |
|---|---|---|---|---|---|
| IsoForest | 0.056 | 0.041 | 0.222 | 0.070 | 171.0 |
| KNN-Dist | 0.301 | 0.161 | 0.326 | 0.546 | 63.5 |
| Obs-MAD | 0.316 | 0.142 | 0.290 | 0.552 | 117.6 |
| PCA-LinPred-LN | 0.186 | 0.104 | 0.244 | 0.320 | 137.5 |
| PCA-Recon | 0.312 | 0.198 | 0.331 | 0.584 | 46.2 |
| TS-JEPA-NoGate | 0.080 | 0.235 | 0.324 | 0.499 | 112.6 |
| ZScoreMax | 0.339 | 0.209 | 0.374 | 0.609 | 67.5 |

TABLE S24
(A) FRACTION OF STREAM-SEED CELLS WHERE EACH MODEL'S POINT-F1 EXCEEDS THE TRIVIAL ALL-POSITIVE F1 $2\pi/(1+\pi)$ AT $q^*$=0.05; (B) MEAN DETECTION DELAY ON THE INTERSECTION OF EVENTS DETECTED BY BOTH THE MODEL AND TS-JEPA (STEPS).

| Model | $P(\text{F1} > \text{trivial})$ | Point-F1 | $\Delta t$ model | $\Delta t$ TS-JEPA |
|---|---|---|---|---|
| JEPA-CtxRecon | 0.30 | 0.190 | 40.3 | 46.5 |
| JEPA-NoEMA | 0.33 | 0.200 | 49.4 | 46.9 |
| OpEntropy | 0.37 | 0.204 | 42.3 | 43.9 |
| PCA-Recon | 0.50 | 0.213 | 49.6 | 59.0 |
| PotentialFlow | 0.18 | 0.106 | 27.7 | 45.7 |
| Reynolds | 0.30 | 0.178 | 58.6 | 45.1 |
| TCN-ObsPred | 0.39 | 0.158 | 51.3 | 60.3 |
| TCN-ObsRecon | 0.36 | 0.150 | 50.7 | 60.0 |
| TS-JEPA | 0.31 | 0.197 | – | – |
| TimesNet | 0.38 | 0.216 | 56.9 | 59.2 |
| TranAD | 0.24 | 0.148 | 120.9 | 60.3 |
| ZScoreMax | 0.39 | 0.217 | 90.3 | 59.8 |

TABLE S25
SHARED INTERSECTING EVENT DETECTION DELAY ($q^*$=0.05). PAIRED RUNS: STREAM–SEED CELLS WHERE BOTH MODELS DETECTED AT LEAST ONE COMMON EVENT; SHARED EVENTS: TOTAL INTERSECTING EVENTS; $\Delta t_{\text{TS-JEPA}}$ AND $\Delta t_{\text{Model}}$: MEAN DETECTION DELAY OVER SHARED EVENTS (STEPS); DIFFERENCE: $\Delta t_{\text{TS-JEPA}} - \Delta t_{\text{Model}}$.

| **Model Contrast** | **Paired Runs** | **Shared Events** | $\mathbf{\Delta t_{\text{TS-JEPA}}}$ | $\mathbf{\Delta t_{\text{Model}}}$ | **Difference** |
|---|---|---|---|---|---|
| JEPA-CtxRecon | 78 | 404 | 46.52 | 40.30 | +6.22 |
| JEPA-NoEMA | 79 | 578 | 46.87 | 49.38 | -2.51 |
| OpEntropy | 85 | 697 | 43.86 | 42.27 | +1.59 |
| PCA-Recon | 65 | 243 | 59.01 | 49.58 | +9.43 |
| PotentialFlow | 77 | 513 | 45.70 | 27.68 | +18.02 |
| Reynolds | 77 | 827 | 45.11 | 58.60 | -13.49 |
| TCN-ObsPred | 65 | 249 | 60.35 | 51.33 | +9.01 |
| TCN-ObsRecon | 65 | 242 | 60.04 | 50.66 | +9.38 |
| TimesNet | 65 | 244 | 59.21 | 56.91 | +2.30 |
| TranAD | 63 | 230 | 60.27 | 120.92 | -60.65 |
| ZScoreMax | 69 | 252 | 59.79 | 90.32 | -30.53 |

TABLE S26
STREAM-LEVEL DRIVERS OF REALIZED SPOT FPR ACROSS ALL DUMPED CELLS (SPEARMAN $\rho$): CALIBRATION→TEST KS DRIFT PREDICTS FAILURE; CALIBRATION-BLOCK KURTOSIS IS MILDLY PROTECTIVE.

| Predictor | $\rho$ vs. realized FPR | $p$ |
|---|---|---|
| KS drift (cal → test) | 0.377 | 7.4e-38 |
| Cal-block excess kurtosis | -0.315 | 2.3e-26 |

TABLE S27
POTENTIAL-FLOW INFERENCE-PATH TEST: SCORING AT THE $z_{\text{NOISE}}$=0 MEAN PATH VS. NOISE-MATCHED QUERYING ($z$=0.5($z_{\text{NOISE}}$+$z_{\text{TGT}}$), 16 DRAWS). NOISE MATCHING DOES NOT RESCUE PERFORMANCE, REJECTING THE TRAIN/TEST-MISMATCH EXPLANATION.

| Stream | Mapping | Path | FPR | Event Recall |
|---|---|---|---|---|
| Daphnet:S01R01E1 | smear | mean ($z_n$=0) | 0.055 | 0.071 |
| Daphnet:S01R01E1 | trailing | mean ($z_n$=0) | 0.031 | 0.714 |
| Daphnet:S01R01E1 | smear | noise-matched | 0.016 | 0.000 |
| Daphnet:S01R01E1 | trailing | noise-matched | 0.010 | 0.214 |
| SMAP:P-3 | smear | mean ($z_n$=0) | 0.000 | 0.000 |
| SMAP:P-3 | trailing | mean ($z_n$=0) | 0.002 | 1.000 |
| SMAP:P-3 | smear | noise-matched | 0.000 | 0.000 |
| SMAP:P-3 | trailing | noise-matched | 0.001 | 1.000 |
| swan:sf | smear | mean ($z_n$=0) | 0.009 | 0.011 |
| swan:sf | trailing | mean ($z_n$=0) | 0.001 | 0.002 |
| swan:sf | smear | noise-matched | 0.008 | 0.015 |
| swan:sf | trailing | noise-matched | 0.002 | 0.004 |

TABLE S28
MULTI-DATASET BENCHMARK COMPARISON FOR PUBLISHED DEEP ANOMALY DETECTION BASELINES ACROSS 11 TELEMETRY DATASETS (130 STREAMS; UNIFIED PIPELINE UNDER SPOT CALIBRATION). PANEL A REPORTS UNADJUSTED POINT-F1; PANEL B REPORTS POINT-ADJUSTED F1 (PA-F1). MACRO AVERAGES WEIGHT EACH DATASET EQUALLY; STREAM-MACRO AVERAGES ACROSS ALL 130 EVALUATION STREAMS. BEST RESULT IN **BOLD**, SECOND UNDERLINED.

| Dataset | Streams | NCAD | DCdetector | Anom.Trans. | TimesNet | TranAD | TS-JEPA |
|---|---|---|---|---|---|---|---|
| *Panel A: Unadjusted Point-F1* | | | | | | | |
| CalIt2 | 1 | 0.019 | 0.049 | 0.122 | 0.147 | **0.165** | 0.047 |
| CICIDS | 6 | 0.010 | 0.004 | 0.011 | 0.091 | **0.118** | 0.035 |
| Daphnet | 26 | 0.080 | 0.124 | 0.104 | **0.141** | 0.120 | 0.099 |
| GECCO | 1 | 0.192 | **0.216** | 0.050 | 0.030 | 0.030 | 0.205 |
| Genesis | 1 | 0.000 | 0.000 | 0.070 | 0.000 | 0.000 | **0.077** |
| GHL | 14 | 0.000 | 0.000 | 0.016 | 0.025 | 0.015 | **0.048** |
| Metro | 1 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | **0.001** |
| MSL | 26 | 0.073 | 0.070 | 0.101 | 0.181 | 0.205 | **0.217** |
| Room-Occ. | 2 | 0.055 | 0.149 | 0.355 | 0.540 | **0.543** | 0.355 |
| SMAP | 51 | 0.137 | 0.136 | 0.079 | 0.177 | 0.171 | **0.267** |
| SWAN | 1 | 0.070 | 0.020 | 0.097 | 0.082 | 0.348 | **0.404** |
| **Macro Mean** | 11 | 0.058 | 0.070 | 0.091 | 0.129 | 0.156 | **0.160** |
| **Stream Mean** | 130 | 0.088 | 0.097 | 0.082 | 0.151 | 0.152 | **0.186** |
| *Panel B: Point-Adjusted F1 (PA-F1)* | | | | | | | |
| CalIt2 | 1 | 0.019 | 0.049 | 0.122 | 0.147 | **0.165** | 0.077 |
| CICIDS | 6 | 0.077 | 0.004 | 0.044 | 0.102 | **0.222** | 0.170 |
| Daphnet | 26 | 0.412 | 0.441 | **0.472** | 0.343 | 0.309 | 0.421 |
| GECCO | 1 | **0.507** | 0.319 | 0.056 | 0.030 | 0.030 | 0.456 |
| Genesis | 1 | 0.000 | 0.000 | 0.202 | 0.000 | 0.000 | **0.297** |
| GHL | 14 | 0.000 | 0.000 | **0.219** | 0.032 | 0.088 | 0.089 |
| Metro | 1 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | **0.001** |
| MSL | 26 | 0.212 | 0.263 | 0.247 | 0.364 | 0.377 | **0.500** |
| Room-Occ. | 2 | 0.426 | 0.235 | 0.429 | 0.554 | **0.641** | 0.355 |
| SMAP | 51 | 0.390 | 0.293 | 0.159 | 0.281 | 0.227 | **0.467** |
| SWAN | 1 | **0.725** | 0.713 | 0.724 | 0.714 | 0.681 | 0.578 |
| **Macro Mean** | 11 | 0.252 | 0.211 | 0.243 | 0.233 | 0.249 | **0.310** |
| **Stream Mean** | 130 | 0.298 | 0.268 | 0.247 | 0.275 | 0.263 | **0.401** |

TABLE S29

MASTER HEADLINE FALSE-POSITIVE RATE (FPR) RECONCILIATION TABLE. CROSS-REFERENCE MAPPING ALL REPORTED FPR VALUES ACROSS THE MANUSCRIPT AND SUPPLEMENTARY MATERIAL TO THEIR EVALUATION COHORTS, RANDOM SEED COVERAGE, TEMPORAL ATTRIBUTION REGIMES, CALIBRATION CONFIGURATIONS, AGGREGATION LEVELS, AND PRIMARY TEXT LOCATIONS.

| Reported FPR | Model / Architecture | Evaluation Cohort | Seeds | Attribution | Calibrator | Aggregation | Primary Manuscript Context |
|---|---|---|---|---|---|---|---|
| 0.076 (7.6%) | TS-JEPA | 19 non-GHL streams | 5 | Buffered (smear) | SPOT ($u$=0.98, $q$=$10^{-3}$) | Stream-macro | Primary held-out benchmark (Section VII-A, Table II, Table I(a)) |
| 0.300 (30.0%) | TimesNet | 19 non-GHL streams | 5 | Buffered (smear) | SPOT ($u$=0.98, $q$=$10^{-3}$) | Stream-macro | Baseline reconstructive benchmark (Section VII-A, Table II, Table I(a)) |
| 0.307 (30.7%) | TranAD | 19 non-GHL streams | 5 | Buffered (smear) | SPOT ($u$=0.98, $q$=$10^{-3}$) | Stream-macro | Baseline attention reconstructive benchmark (Section VII-A, Table II, Table I(a)) |
| 0.005 (0.5%) | Linear AR Forecaster | 19 non-GHL streams | 5 | Buffered (smear) | SPOT ($u$=0.98, $q$=$10^{-3}$) | Stream-macro | Linear dynamical control (Section VII-A) |
| 0.0526 (5.26%) | TS-JEPA | 33 primary streams | 1–5 | Buffered (smear) | Static SPOT ($u$=0.98, $q$=$10^{-3}$) | Stream-macro | 33-stream calibrator comparison (Section VII-D, Table III) |
| 0.2217 (22.17%) | TimesNet / TranAD | 33 primary streams | 1–5 | Buffered (smear) | Static SPOT ($u$=0.98, $q$=$10^{-3}$) | Stream-macro | 33-stream calibrator comparison (Section VII-D, Table III) |
| 0.080 (8.0%) | TS-JEPA | 19 non-GHL streams | 5 | Buffered (smear) | SPOT ($u$=0.98, $q$=$10^{-3}$) | Stream-macro | Experiment A controlled backbone study (Section VI-E, Table S19) |
| 0.186 (18.6%) | TCN-ObsPred-LN / PCA-LinPred-LN | 19 non-GHL streams | 5 | Buffered (smear) | SPOT ($u$=0.98, $q$=$10^{-3}$) | Stream-macro | Experiment A bounded score controls (Section VI-E, Table S19) |
| 0.014 (1.4%) | TS-JEPA | 19 non-GHL streams | 5 | Causal (trailing) | SPOT ($u$=0.98, $q$=$10^{-3}$) | Stream-macro | Strictly causal streaming benchmark (Section VI-E, Table S31) |
| 0.118 (11.8%) | TS-JEPA | 15-stream matched backbone | 1 | Buffered (smear) | SPOT ($u$=0.98, $q$=$10^{-3}$) | Stream-macro | Preliminary 15-stream noise robustness ablation (Section VI-E, Table S2) |
| 0.33% (0.0033) | TS-JEPA | 6-stream pilot diagnostic | 1 | Buffered (smear) | SPOT ($u$=0.98, $q$=$10^{-3}$) | Stream-macro | EVT failure-mechanism decomposition (Section S-6, Table S14) |

TABLE S30

COMPLETE 17-MODEL COMMON SUPPORT MATRIX ON REALIZED FPR $[0.10, 0.50]$ (STRICTLY CAUSAL TRAILING REGIME). FOR EACH MODEL CONTRASTED AGAINST TS-JEPA, COMMON SUPPORT REQUIRES OBSERVED TEST FALSE-POSITIVE RATES TO SPAN $[0.10, 0.50]$ WITHOUT EXTRAPOLATION. SUPPORTED AND DISCARDED STREAMS ARE OUT OF 19 TOTAL NON-GHL BENCHMARK STREAMS. SEED CELLS INDICATE PAIRED STREAM–SEED RUNS. $\Delta$pAUC: MEAN PAIRED PARTIAL-AUC DIFFERENCE (TS-JEPA MINUS MODEL) ACROSS SUPPORTED STREAMS; 95% CI: STREAM-LEVEL BOOTSTRAP CONFIDENCE INTERVAL (2,000 RESAMPLES); WILCOXON $p$: TWO-SIDED STREAM-LEVEL PAIRED WILCOXON SIGNED-RANK TEST; FDR $p_{\mathrm{BH}}$: BENJAMINI–HOCHBERG ADJUSTED $p$-VALUE ACROSS THE 17-MODEL FAMILY.

| Model Contrast (vs. TS-JEPA) | Supported Streams | Discarded Streams | Seed Cells | Datasets | $\Delta$pAUC | 95% Bootstrap CI | Wilcoxon $p$ | FDR $p_{\mathrm{BH}}$ |
|---|---|---|---|---|---|---|---|---|
| IsoForest | 11 | 8 | 54 | 6 | +0.025 | [-0.077, +0.143] | 0.844 | 1.0 |
| JEPA-CtxRecon | 13 | 6 | 53 | 6 | -0.007 | [-0.018, +0.005] | 0.312 | 1.0 |
| JEPA-NoEMA | 14 | 5 | 56 | 6 | +0.011 | [-0.020, +0.050] | 1.000 | 1.0 |
| KNN-Dist | 5 | 14 | 22 | 4 | +0.094 | [-0.036, +0.270] | 0.438 | 1.0 |
| Obs-MAD | 5 | 14 | 22 | 4 | +0.028 | [-0.055, +0.146] | 1.000 | 1.0 |
| OpEntropy | 13 | 6 | 56 | 6 | -0.012 | [-0.037, +0.005] | 0.312 | 1.0 |
| PCA-LinPred-LN | 11 | 8 | 54 | 5 | +0.121 | [-0.008, +0.271] | 0.312 | 1.0 |
| PCA-Recon | 5 | 14 | 22 | 4 | +0.126 | [-0.028, +0.307] | 0.438 | 1.0 |
| PotentialFlow | 14 | 5 | 62 | 6 | -0.017 | [-0.059, +0.009] | 0.499 | 1.0 |
| Reynolds | 13 | 6 | 55 | 6 | +0.002 | [-0.047, +0.037] | 0.297 | 1.0 |
| TCN-ObsPred | 4 | 15 | 20 | 4 | +0.106 | [-0.057, +0.326] | 0.625 | 1.0 |
| TCN-ObsPred-LN | 9 | 10 | 42 | 6 | -0.018 | [-0.044, +0.002] | 0.219 | 1.0 |
| TCN-ObsRecon | 4 | 15 | 20 | 4 | +0.130 | [-0.040, +0.349] | 0.375 | 1.0 |
| TS-JEPA-NoGate | 14 | 5 | 61 | 6 | +0.005 | [-0.012, +0.023] | 0.674 | 1.0 |
| TimesNet | 4 | 15 | 20 | 4 | +0.061 | [-0.153, +0.329] | 0.625 | 1.0 |
| TranAD | 5 | 14 | 21 | 4 | +0.085 | [-0.058, +0.289] | 0.812 | 1.0 |
| ZScoreMax | 5 | 14 | 25 | 4 | +0.032 | [-0.049, +0.159] | 0.875 | 1.0 |

TABLE S31

STRICTLY CAUSAL STREAMING BENCHMARK (0-LOOKAHEAD, TRAILING ATTRIBUTION), 33 STREAMS, HELD-OUT SPOT ($q$=$10^{-3}$). "NON-GHL": 19 STREAMS AT FIVE SEEDS (STREAM-MACRO, $\pm$ SEED SD); "GHL": 14 STREAMS AT THREE-TO-FIVE SEEDS. LATENT PREDICTORS CUT NOMINAL FALSE ALARMS TO 0.97%–1.89% ON NON-GHL (12–25$\times$ BELOW RECONSTRUCTIVE BASELINES) AT THE COST OF LOWER CAUSAL POINT-F1 AND EVENT RECALL.

| | Non-GHL Streams (19 Streams, 5 Seeds) | | | | | Gasoil Heating Loop (14 Streams) | | |
|---|---|---|---|---|---|---|---|---|
| Model Architecture | Held-Out FPR ↓ | Point-F1 (Mean $\pm$ SD) ↑ | PR-AUC ↑ | Event Recall ↑ | Test FPR ↓ | Held-Out FPR ↓ | Point-F1 ↑ | Event Recall ↑ |
| *Latent-Space Predictive Architectures (JEPAs)* | | | | | | | | |
| TS-JEPA (Unconstrained Control) | 0.0136 | 0.1118 $\pm$ 0.0412 | 0.2672 | 0.4655 | 0.0844 | 0.0035 | 0.000 | 0.000 |
| Operator-Entropy JEPA | 0.0189 | 0.1093 $\pm$ 0.0085 | 0.2528 | 0.4692 | 0.0986 | 0.0085 | 0.001 | 0.024 |
| Reynolds-Stress JEPA | <u>0.0111</u> | 0.0876 $\pm$ 0.0367 | <u>0.2700</u> | 0.4563 | <u>0.0842</u> | **0.0023** | 0.002 | 0.014 |
| Potential-Flow JEPA | **0.0097** | 0.0233 $\pm$ 0.0166 | 0.1775 | 0.3428 | **0.0289** | 0.0047 | <u>0.006</u> | <u>0.032</u> |
| *Observation-Space Reconstructive Baselines* | | | | | | | | |
| TimesNet (ICLR 2023) | 0.2427 | **0.2091 $\pm$ 0.0265** | **0.2931** | **0.5882** | 0.2163 | 0.0563 | **0.022** | **0.850** |
| TranAD (VLDB 2022) | 0.2219 | <u>0.1405 $\pm$ 0.0154</u> | 0.2483 | <u>0.5334</u> | 0.2027 | 0.0398 | 0.018 | 0.811 |

## S-9. Complete Statistical Test Map

Table S32 enumerates every paired stream-level comparison reported in the manuscript, resolving Reviewer Concern 5. Two cohorts are reported separately: the 19 non-GHL streams evaluated at five seeds (all paired-inference claims) and the full 33-stream primary benchmark (headline means cited in the abstract). Multiple seeds are averaged within a stream before pairing; the independent statistical unit is the telemetry stream. $p_{\mathrm{adj}}$ applies Benjamini–Hochberg FDR control within each cohort $\times$ regime family (45 tests per family). The effect column reports the median paired difference with a 95% bootstrap CI over the stream-level differences; $r_b$ is the rank-biserial correlation.

TABLE S32: Complete statistical test map for paired stream-level comparisons (Wilcoxon signed-rank on stream-macro means, seeds averaged within stream).

| Cohort | Regime | Endpoint | Comparison | $n$ | $p$ | $p_{\mathrm{adj}}$ | Median diff. [CI] | $r_b$ |
|---|---|---|---|---|---|---|---|---|
| non-GHL (n=19) | Buffered | Held-out nominal FPR | TS-JEPA - TimesNet | 19 | 0.0401 | 0.2478 | -0.0713 [-0.3759, -0.0833] | -0.16 |
| | | Held-out nominal FPR | TS-JEPA - TranAD | 19 | 0.0258 | 0.2478 | -0.1511 [-0.3714, -0.0942] | -0.16 |
| | | Held-out nominal FPR | OpEntropy - TS-JEPA | 19 | 0.9843 | 0.9843 | -0.0050 [-0.0278, 0.0485] | -0.16 |
| | | Held-out nominal FPR | Reynolds - TS-JEPA | 19 | 0.9217 | 0.9427 | 0.0004 [-0.0380, 0.0453] | 0.05 |
| | | Held-out nominal FPR | PotentialFlow - TS-JEPA | 19 | 0.7381 | 0.9427 | 0.0017 [-0.0259, 0.0556] | 0.05 |
| | | Held-out nominal FPR | OpEntropy - TimesNet | 19 | 0.0352 | 0.2478 | -0.0566 [-0.3646, -0.0843] | -0.18 |
| | | Held-out nominal FPR | Reynolds - TimesNet | 19 | 0.0198 | 0.2478 | -0.0576 [-0.3536, -0.0939] | -0.22 |
| | | Held-out nominal FPR | PotentialFlow - TimesNet | 19 | 0.0494 | 0.2478 | -0.0857 [-0.3577, -0.0625] | -0.16 |
| | | Held-out nominal FPR | TimesNet - TranAD | 19 | 0.8139 | 0.9427 | 0.0000 [-0.1179, 0.0986] | 0.00 |
| | | Point-F1 | TS-JEPA - TimesNet | 19 | 0.5678 | 0.9162 | 0.0168 [-0.0702, 0.1451] | 0.05 |
| | | Point-F1 | TS-JEPA - TranAD | 19 | 0.2145 | 0.5081 | 0.0168 [-0.0094, 0.2289] | 0.11 |
| | | Point-F1 | OpEntropy - TS-JEPA | 19 | 0.6791 | 0.9427 | -0.0093 [-0.0428, 0.0512] | -0.22 |
| | | Point-F1 | Reynolds - TS-JEPA | 19 | 0.4413 | 0.7944 | -0.0019 [-0.0754, 0.0125] | -0.05 |
| | | Point-F1 | PotentialFlow - TS-JEPA | 19 | 0.0108 | 0.2478 | -0.0946 [-0.2712, -0.0478] | -0.47 |
| | | Point-F1 | OpEntropy - TimesNet | 19 | 0.2579 | 0.5527 | 0.0283 [-0.0679, 0.1416] | 0.26 |
| | | Point-F1 | Reynolds - TimesNet | 19 | 0.8789 | 0.9427 | 0.0000 [-0.0971, 0.1160] | 0.00 |
| | | Point-F1 | PotentialFlow - TimesNet | 19 | 0.1447 | 0.4069 | -0.0295 [-0.2327, -0.0070] | -0.16 |
| | | Point-F1 | TimesNet - TranAD | 19 | 0.0640 | 0.2478 | 0.0000 [0.0180, 0.1303] | 0.29 |
| | | PR-AUC | TS-JEPA - TimesNet | 19 | 0.9217 | 0.9427 | -0.0023 [-0.1733, 0.1570] | -0.16 |
| | | PR-AUC | TS-JEPA - TranAD | 19 | 0.6794 | 0.9427 | 0.0015 [-0.1254, 0.1996] | 0.05 |
| | | PR-AUC | OpEntropy - TS-JEPA | 19 | 0.8288 | 0.9427 | -0.0040 [-0.0357, 0.0362] | -0.26 |
| | | PR-AUC | Reynolds - TS-JEPA | 19 | 0.1232 | 0.3696 | -0.0177 [-0.0445, 0.0152] | -0.26 |
| | | PR-AUC | PotentialFlow - TS-JEPA | 19 | 0.1134 | 0.3644 | -0.0090 [-0.2182, -0.0297] | -0.26 |
| | | PR-AUC | OpEntropy - TimesNet | 19 | 0.9217 | 0.9427 | -0.0003 [-0.1748, 0.1464] | -0.05 |
| | | PR-AUC | Reynolds - TimesNet | 19 | 0.8288 | 0.9427 | -0.0142 [-0.1845, 0.1352] | -0.16 |
| | | PR-AUC | PotentialFlow - TimesNet | 19 | 0.2101 | 0.5081 | -0.0134 [-0.2620, -0.0101] | -0.16 |
| | | PR-AUC | TimesNet - TranAD | 19 | 0.0361 | 0.2478 | 0.0116 [-0.0207, 0.1166] | 0.37 |
| | | Event recall | TS-JEPA - TimesNet | 19 | 0.3938 | 0.7383 | -0.0080 [-0.3681, 0.1526] | -0.18 |
| | | Event recall | TS-JEPA - TranAD | 19 | 0.9056 | 0.9427 | 0.0000 [-0.2633, 0.2440] | 0.06 |
| | | Event recall | OpEntropy - TS-JEPA | 19 | 0.8017 | 0.9427 | 0.0000 [-0.0525, 0.1148] | -0.14 |
| | | Event recall | Reynolds - TS-JEPA | 19 | 0.3937 | 0.7383 | 0.0000 [-0.1710, 0.0905] | -0.06 |
| | | Event recall | PotentialFlow - TS-JEPA | 19 | 0.0533 | 0.2478 | 0.0000 [-0.3347, -0.0341] | -0.20 |
| | | Event recall | OpEntropy - TimesNet | 19 | 0.6050 | 0.9388 | 0.0000 [-0.3026, 0.1440] | 0.00 |
| | | Event recall | Reynolds - TimesNet | 19 | 0.1863 | 0.4931 | -0.0667 [-0.3547, 0.0734] | -0.25 |
| | | Event recall | PotentialFlow - TimesNet | 19 | 0.0610 | 0.2478 | -0.4000 [-0.5145, -0.0525] | -0.16 |
| | | Event recall | TimesNet - TranAD | 19 | 0.0661 | 0.2478 | 0.0000 [0.0022, 0.2007] | 0.56 |
| | | Detection delay (steps) | TS-JEPA - TimesNet | 11 | 0.2783 | 0.5693 | 28.4000 [-36.5661, 67.4297] | 0.45 |
| | | Detection delay (steps) | TS-JEPA - TranAD | 10 | 0.6953 | 0.9427 | 32.1970 [-626.3884, 52.6033] | 0.40 |
| | | Detection delay (steps) | OpEntropy - TS-JEPA | 16 | 0.5701 | 0.9162 | 4.5180 [-21.4413, 347.5172] | 0.20 |
| | | Detection delay (steps) | Reynolds - TS-JEPA | 14 | 0.6698 | 0.9427 | 3.9517 [-15.1168, 105.9640] | 0.00 |
| | | Detection delay (steps) | PotentialFlow - TS-JEPA | 15 | 0.8753 | 0.9427 | -2.0000 [-22.4649, 38.7592] | -0.14 |
| | | Detection delay (steps) | OpEntropy - TimesNet | 13 | 0.0942 | 0.3262 | 39.9176 [-3.4980, 116.3097] | 0.38 |
| | | Detection delay (steps) | Reynolds - TimesNet | 13 | 0.0210 | 0.2478 | 67.0000 [33.0242, 183.0271] | 0.50 |
| | | Detection delay (steps) | PotentialFlow - TimesNet | 13 | 0.2334 | 0.5251 | 43.8857 [-19.9464, 330.3680] | 0.50 |
| | | Detection delay (steps) | TimesNet - TranAD | 12 | 0.4961 | 0.8586 | 0.0000 [-534.4812, 9.8618] | -0.11 |
| non-GHL (n=19) | Causal | Held-out nominal FPR | TS-JEPA - TimesNet | 18 | 0.0086 | 0.0669 | -0.1426 [-0.3580, -0.1185] | -0.29 |
| | | Held-out nominal FPR | TS-JEPA - TranAD | 18 | 0.0086 | 0.0669 | -0.1685 [-0.3271, -0.1053] | -0.29 |
| | | Held-out nominal FPR | OpEntropy - TS-JEPA | 18 | 0.7960 | 0.8330 | 0.0004 [-0.0044, 0.0210] | 0.12 |
| | | Held-out nominal FPR | Reynolds - TS-JEPA | 18 | 0.5862 | 0.7834 | -0.0002 [-0.0103, 0.0039] | -0.06 |
| | | Held-out nominal FPR | PotentialFlow - TS-JEPA | 18 | 0.3692 | 0.5729 | -0.0006 [-0.0140, 0.0036] | -0.33 |
| | | Held-out nominal FPR | OpEntropy - TimesNet | 18 | 0.0086 | 0.0669 | -0.1386 [-0.3531, -0.1135] | -0.29 |
| | | Held-out nominal FPR | Reynolds - TimesNet | 18 | 0.0104 | 0.0669 | -0.1263 [-0.3537, -0.1158] | -0.22 |
| | | Held-out nominal FPR | PotentialFlow - TimesNet | 18 | 0.0104 | 0.0669 | -0.1409 [-0.3593, -0.1155] | -0.22 |
| | | Held-out nominal FPR | TimesNet - TranAD | 18 | 0.7671 | 0.8219 | 0.0000 [-0.0341, 0.1041] | -0.11 |
| | | Point-F1 | TS-JEPA - TimesNet | 18 | 0.1187 | 0.2420 | -0.0336 [-0.1980, -0.0031] | -0.22 |
| | | Point-F1 | TS-JEPA - TranAD | 18 | 0.6397 | 0.7834 | -0.0129 [-0.1352, 0.0725] | -0.11 |
| | | Point-F1 | OpEntropy - TS-JEPA | 18 | 0.5226 | 0.7586 | -0.0027 [-0.0316, 0.0277] | -0.11 |
| | | Point-F1 | Reynolds - TS-JEPA | 18 | 0.0898 | 0.2376 | -0.0081 [-0.0609, 0.0097] | -0.44 |
| | | Point-F1 | PotentialFlow - TS-JEPA | 18 | 0.0090 | 0.0669 | -0.0146 [-0.1523, -0.0349] | -0.44 |
| | | Point-F1 | OpEntropy - TimesNet | 18 | 0.0840 | 0.2363 | -0.0252 [-0.1886, -0.0111] | -0.29 |
| | | Point-F1 | Reynolds - TimesNet | 18 | 0.0245 | 0.1104 | -0.0842 [-0.2134, -0.0354] | -0.41 |
| | | Point-F1 | PotentialFlow - TimesNet | 18 | 0.0015 | 0.0661 | -0.0801 [-0.3002, -0.0854] | -0.73 |
| | | Point-F1 | TimesNet - TranAD | 18 | 0.0392 | 0.1472 | 0.0008 [0.0136, 0.1444] | 0.38 |
| | | PR-AUC | TS-JEPA - TimesNet | 18 | 0.7660 | 0.8219 | -0.0052 [-0.1437, 0.0892] | -0.11 |
| | | PR-AUC | TS-JEPA - TranAD | 18 | 0.7019 | 0.7896 | 0.0093 [-0.0953, 0.1395] | 0.11 |
| | | PR-AUC | OpEntropy - TS-JEPA | 18 | 0.0599 | 0.1927 | -0.0050 [-0.0436, 0.0167] | -0.56 |
| | | PR-AUC | Reynolds - TS-JEPA | 18 | 0.1964 | 0.3535 | -0.0054 [-0.0295, 0.0425] | -0.33 |

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| Cohort | Regime | Endpoint | Comparison | $n$ | $p$ | $p_{\mathrm{adj}}$ | Median diff. [CI] | $r_b$ |
|---|---|---|---|---|---|---|---|---|
| | | PR-AUC | PotentialFlow - TS-JEPA | 18 | 0.4171 | 0.6257 | -0.0010 [-0.1788, -0.0127] | 0.00 |
| | | PR-AUC | OpEntropy - TimesNet | 18 | 0.6397 | 0.7834 | -0.0010 [-0.1547, 0.0666] | -0.11 |
| | | PR-AUC | Reynolds - TimesNet | 18 | 0.6095 | 0.7834 | -0.0046 [-0.1471, 0.0947] | -0.22 |
| | | PR-AUC | PotentialFlow - TimesNet | 18 | 0.2462 | 0.4261 | -0.0035 [-0.2195, -0.0287] | 0.00 |
| | | PR-AUC | TimesNet - TranAD | 18 | 0.0304 | 0.1242 | 0.0086 [-0.0201, 0.1242] | 0.44 |
| | | Event recall | TS-JEPA - TimesNet | 18 | 0.2656 | 0.4427 | -0.1000 [-0.2926, 0.0496] | -0.06 |
| | | Event recall | TS-JEPA - TranAD | 18 | 0.5418 | 0.7619 | -0.0889 [-0.2420, 0.1024] | 0.00 |
| | | Event recall | OpEntropy - TS-JEPA | 18 | 0.9176 | 0.9384 | -0.0040 [-0.0920, 0.0954] | -0.12 |
| | | Event recall | Reynolds - TS-JEPA | 18 | 0.9645 | 0.9645 | 0.0000 [-0.1117, 0.0853] | -0.09 |
| | | Event recall | PotentialFlow - TS-JEPA | 18 | 0.1237 | 0.2420 | -0.1000 [-0.2694, 0.0323] | -0.53 |
| | | Event recall | OpEntropy - TimesNet | 18 | 0.2802 | 0.4504 | 0.0000 [-0.3077, 0.0661] | -0.07 |
| | | Event recall | Reynolds - TimesNet | 18 | 0.1053 | 0.2420 | -0.2000 [-0.2770, 0.0233] | -0.33 |
| | | Event recall | PotentialFlow - TimesNet | 18 | 0.0156 | 0.0781 | -0.0962 [-0.4292, -0.0750] | -0.43 |
| | | Event recall | TimesNet - TranAD | 18 | 0.1159 | 0.2420 | 0.0000 [-0.0125, 0.1420] | 0.67 |
| | | Detection delay (steps) | TS-JEPA - TimesNet | 13 | 0.0681 | 0.2043 | 45.6000 [-2.1414, 137.9843] | 0.54 |
| | | Detection delay (steps) | TS-JEPA - TranAD | 13 | 0.1465 | 0.2747 | 58.8000 [-468.1920, 132.5104] | 0.69 |
| | | Detection delay (steps) | OpEntropy - TS-JEPA | 17 | 0.6441 | 0.7834 | 1.1667 [-38.0739, 25.5760] | 0.06 |
| | | Detection delay (steps) | Reynolds - TS-JEPA | 17 | 0.0980 | 0.2420 | 16.4667 [7.1296, 103.0613] | 0.38 |
| | | Detection delay (steps) | PotentialFlow - TS-JEPA | 14 | 0.6698 | 0.7896 | -7.6306 [-51.3326, 48.5405] | -0.14 |
| | | Detection delay (steps) | OpEntropy - TimesNet | 13 | 0.0479 | 0.1656 | 75.0000 [-4.1865, 117.4269] | 0.54 |
| | | Detection delay (steps) | Reynolds - TimesNet | 13 | 0.0134 | 0.0755 | 39.9631 [36.8358, 234.6891] | 0.69 |
| | | Detection delay (steps) | PotentialFlow - TimesNet | 12 | 0.1099 | 0.2420 | 68.7500 [0.8355, 113.9374] | 0.50 |
| | | Detection delay (steps) | TimesNet - TranAD | 13 | 0.6953 | 0.7896 | 0.0000 [-495.3941, 16.5770] | -0.20 |
| primary (n=33) | Buffered | Held-out nominal FPR | TS-JEPA - TimesNet | 33 | 0.0024 | 0.0179 | -0.0378 [-0.2454, -0.0665] | -0.33 |
| | | Held-out nominal FPR | TS-JEPA - TranAD | 33 | 0.0029 | 0.0187 | -0.0636 [-0.2439, -0.0670] | -0.33 |
| | | Held-out nominal FPR | OpEntropy - TS-JEPA | 33 | 0.7509 | 0.8604 | -0.0007 [-0.0131, 0.0316] | -0.03 |
| | | Held-out nominal FPR | Reynolds - TS-JEPA | 33 | 0.9158 | 0.9158 | -0.0008 [-0.0229, 0.0249] | -0.03 |
| | | Held-out nominal FPR | PotentialFlow - TS-JEPA | 33 | 0.3215 | 0.5358 | 0.0027 [-0.0130, 0.0341] | 0.27 |
| | | Held-out nominal FPR | OpEntropy - TimesNet | 33 | 0.0061 | 0.0304 | -0.0514 [-0.2404, -0.0621] | -0.29 |
| | | Held-out nominal FPR | Reynolds - TimesNet | 33 | 0.0011 | 0.0103 | -0.0534 [-0.2417, -0.0742] | -0.44 |
| | | Held-out nominal FPR | PotentialFlow - TimesNet | 33 | 0.0078 | 0.0352 | -0.0562 [-0.2361, -0.0539] | -0.27 |
| | | Held-out nominal FPR | TimesNet - TranAD | 33 | 0.2641 | 0.4755 | 0.0000 [-0.0654, 0.0671] | 0.28 |
| | | Point-F1 | TS-JEPA - TimesNet | 33 | 0.5372 | 0.8058 | -0.0044 [-0.0497, 0.0764] | -0.38 |
| | | Point-F1 | TS-JEPA - TranAD | 33 | 0.7648 | 0.8604 | -0.0050 [-0.0138, 0.1286] | -0.38 |
| | | Point-F1 | OpEntropy - TS-JEPA | 33 | 0.7089 | 0.8577 | 0.0000 [-0.0245, 0.0305] | -0.10 |
| | | Point-F1 | Reynolds - TS-JEPA | 33 | 0.6143 | 0.8134 | 0.0000 [-0.0431, 0.0079] | 0.05 |
| | | Point-F1 | PotentialFlow - TS-JEPA | 33 | 0.0325 | 0.1180 | 0.0000 [-0.1581, -0.0213] | -0.33 |
| | | Point-F1 | OpEntropy - TimesNet | 33 | 0.9107 | 0.9158 | -0.0041 [-0.0469, 0.0731] | -0.25 |
| | | Point-F1 | Reynolds - TimesNet | 33 | 0.2557 | 0.4755 | -0.0045 [-0.0656, 0.0588] | -0.35 |
| | | Point-F1 | PotentialFlow - TimesNet | 33 | 0.0387 | 0.1245 | -0.0097 [-0.1456, -0.0087] | -0.39 |
| | | Point-F1 | TimesNet - TranAD | 33 | 0.1271 | 0.3443 | 0.0000 [0.0096, 0.0785] | 0.07 |
| | | PR-AUC | TS-JEPA - TimesNet | 33 | 0.4473 | 0.7034 | 0.0024 [-0.0968, 0.0887] | 0.21 |
| | | PR-AUC | TS-JEPA - TranAD | 33 | 0.2075 | 0.4284 | 0.0024 [-0.0723, 0.1187] | 0.33 |
| | | PR-AUC | OpEntropy - TS-JEPA | 33 | 0.8601 | 0.9158 | -0.0008 [-0.0190, 0.0219] | -0.15 |
| | | PR-AUC | Reynolds - TS-JEPA | 33 | 0.1301 | 0.3443 | -0.0033 [-0.0255, 0.0116] | -0.27 |
| | | PR-AUC | PotentialFlow - TS-JEPA | 33 | 0.5843 | 0.8134 | 0.0002 [-0.1304, -0.0129] | 0.03 |
| | | PR-AUC | OpEntropy - TimesNet | 33 | 0.2565 | 0.4755 | 0.0035 [-0.0944, 0.0876] | 0.21 |
| | | PR-AUC | Reynolds - TimesNet | 33 | 0.7243 | 0.8577 | 0.0011 [-0.1064, 0.0801] | 0.03 |
| | | PR-AUC | PotentialFlow - TimesNet | 33 | 0.8601 | 0.9158 | 0.0022 [-0.1545, 0.0004] | 0.21 |
| | | PR-AUC | TimesNet - TranAD | 33 | 0.1495 | 0.3542 | 0.0003 [-0.0138, 0.0690] | 0.03 |
| | | Event recall | TS-JEPA - TimesNet | 33 | 0.0003 | 0.0044 | -0.8000 [-0.6479, -0.2364] | -0.53 |
| | | Event recall | TS-JEPA - TranAD | 33 | 0.0034 | 0.0190 | -0.5000 [-0.5705, -0.1469] | -0.42 |
| | | Event recall | OpEntropy - TS-JEPA | 33 | 0.4533 | 0.7034 | 0.0000 [-0.0224, 0.0816] | 0.00 |
| | | Event recall | Reynolds - TS-JEPA | 33 | 0.6146 | 0.8134 | 0.0000 [-0.0930, 0.0655] | 0.05 |
| | | Event recall | PotentialFlow - TS-JEPA | 33 | 0.1623 | 0.3653 | 0.0000 [-0.1887, 0.0049] | -0.06 |
| | | Event recall | OpEntropy - TimesNet | 33 | 0.0006 | 0.0070 | -0.6000 [-0.6084, -0.2193] | -0.45 |
| | | Event recall | Reynolds - TimesNet | 33 | 0.0003 | 0.0044 | -0.6000 [-0.6430, -0.2752] | -0.59 |
| | | Event recall | PotentialFlow - TimesNet | 33 | $<$0.0001 | 0.0032 | -0.8000 [-0.7058, -0.3426] | -0.45 |
| | | Event recall | TimesNet - TranAD | 33 | 0.0341 | 0.1180 | 0.0000 [0.0159, 0.1528] | 0.50 |
| | | Detection delay (steps) | TS-JEPA - TimesNet | 11 | 0.2783 | 0.4817 | 28.4000 [-38.1938, 65.4996] | 0.45 |
| | | Detection delay (steps) | TS-JEPA - TranAD | 10 | 0.6953 | 0.8577 | 32.1970 [-627.2807, 54.1011] | 0.40 |
| | | Detection delay (steps) | OpEntropy - TS-JEPA | 16 | 0.5701 | 0.8134 | 4.5180 [-22.0503, 349.4281] | 0.20 |
| | | Detection delay (steps) | Reynolds - TS-JEPA | 14 | 0.6698 | 0.8577 | 3.9517 [-14.5613, 110.2057] | 0.00 |
| | | Detection delay (steps) | PotentialFlow - TS-JEPA | 15 | 0.8753 | 0.9158 | -2.0000 [-22.8252, 37.3496] | -0.14 |
| | | Detection delay (steps) | OpEntropy - TimesNet | 15 | 0.0480 | 0.1440 | 39.9176 [6.2208, 121.1460] | 0.43 |
| | | Detection delay (steps) | Reynolds - TimesNet | 15 | 0.0303 | 0.1180 | 67.0000 [33.7361, 254.5029] | 0.43 |
| | | Detection delay (steps) | PotentialFlow - TimesNet | 14 | 0.2094 | 0.4284 | 35.0845 [-17.2960, 313.0068] | 0.50 |
| | | Detection delay (steps) | TimesNet - TranAD | 25 | 0.1477 | 0.3542 | 0.0000 [-304.7382, 13.9861] | -0.25 |
| primary (n=33) | Causal | Held-out nominal FPR | TS-JEPA - TimesNet | 32 | 0.0002 | 0.0013 | -0.0549 [-0.2354, -0.0804] | -0.42 |
| | | Held-out nominal FPR | TS-JEPA - TranAD | 32 | 0.0002 | 0.0013 | -0.0517 [-0.2094, -0.0707] | -0.42 |
| | | Held-out nominal FPR | OpEntropy - TS-JEPA | 32 | 0.3820 | 0.5372 | 0.0008 [-0.0015, 0.0151] | 0.27 |
| | | Held-out nominal FPR | Reynolds - TS-JEPA | 32 | 0.5052 | 0.6315 | -0.0004 [-0.0059, 0.0018] | -0.10 |
| | | Held-out nominal FPR | PotentialFlow - TS-JEPA | 32 | 0.7190 | 0.7670 | -0.0004 [-0.0074, 0.0029] | -0.12 |
| | | Held-out nominal FPR | OpEntropy - TimesNet | 32 | 0.0004 | 0.0019 | -0.0550 [-0.2243, -0.0791] | -0.35 |
| | | Held-out nominal FPR | Reynolds - TimesNet | 32 | $<$0.0001 | 0.0013 | -0.0558 [-0.2360, -0.0821] | -0.44 |
| | | Held-out nominal FPR | PotentialFlow - TimesNet | 32 | 0.0002 | 0.0013 | -0.0556 [-0.2374, -0.0842] | -0.31 |
| | | Held-out nominal FPR | TimesNet - TranAD | 32 | 0.4264 | 0.5717 | 0.0000 [-0.0132, 0.0664] | 0.27 |
| | | Point-F1 | TS-JEPA - TimesNet | 32 | 0.0135 | 0.0435 | -0.0068 [-0.1229, -0.0087] | -0.55 |
| | | Point-F1 | TS-JEPA - TranAD | 32 | 0.0811 | 0.1826 | -0.0065 [-0.0824, 0.0352] | -0.48 |
| | | Point-F1 | OpEntropy - TS-JEPA | 32 | 0.6580 | 0.7351 | 0.0000 [-0.0183, 0.0165] | -0.05 |
| | | Point-F1 | Reynolds - TS-JEPA | 32 | 0.1590 | 0.2523 | 0.0000 [-0.0333, 0.0064] | -0.37 |

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| Cohort | Regime | Endpoint | Comparison | $n$ | $p$ | $p_{\mathrm{adj}}$ | Median diff. [CI] | $r_b$ |
|---|---|---|---|---|---|---|---|---|
| | | Point-F1 | PotentialFlow - TS-JEPA | 32 | 0.0400 | 0.1126 | 0.0000 [-0.0876, -0.0140] | -0.30 |
| | | Point-F1 | OpEntropy - TimesNet | 32 | 0.0093 | 0.0321 | -0.0074 [-0.1223, -0.0153] | -0.60 |
| | | Point-F1 | Reynolds - TimesNet | 32 | 0.0032 | 0.0119 | -0.0187 [-0.1342, -0.0265] | -0.60 |
| | | Point-F1 | PotentialFlow - TimesNet | 32 | 0.0002 | 0.0013 | -0.0245 [-0.1874, -0.0493] | -0.72 |
| | | Point-F1 | TimesNet - TranAD | 32 | 0.0775 | 0.1826 | 0.0000 [0.0091, 0.0885] | 0.00 |
| | | PR-AUC | TS-JEPA - TimesNet | 32 | 0.4883 | 0.6278 | 0.0020 [-0.0792, 0.0536] | 0.25 |
| | | PR-AUC | TS-JEPA - TranAD | 32 | 0.1195 | 0.2150 | 0.0040 [-0.0508, 0.0793] | 0.38 |
| | | PR-AUC | OpEntropy - TS-JEPA | 32 | 0.1545 | 0.2523 | -0.0013 [-0.0240, 0.0111] | -0.31 |
| | | PR-AUC | Reynolds - TS-JEPA | 32 | 0.2699 | 0.4021 | -0.0030 [-0.0151, 0.0255] | -0.31 |
| | | PR-AUC | PotentialFlow - TS-JEPA | 32 | 0.8034 | 0.8216 | 0.0014 [-0.1044, -0.0039] | 0.19 |
| | | PR-AUC | OpEntropy - TimesNet | 32 | 0.4319 | 0.5717 | 0.0022 [-0.0838, 0.0417] | 0.19 |
| | | PR-AUC | Reynolds - TimesNet | 32 | 0.7329 | 0.7670 | 0.0002 [-0.0800, 0.0580] | 0.00 |
| | | PR-AUC | PotentialFlow - TimesNet | 32 | 0.6247 | 0.7351 | 0.0031 [-0.1216, -0.0081] | 0.31 |
| | | PR-AUC | TimesNet - TranAD | 32 | 0.1195 | 0.2150 | 0.0003 [-0.0116, 0.0681] | 0.06 |
| | | Event recall | TS-JEPA - TimesNet | 32 | 0.0003 | 0.0013 | -0.4462 [-0.6130, -0.2721] | -0.47 |
| | | Event recall | TS-JEPA - TranAD | 32 | 0.0007 | 0.0029 | -0.3667 [-0.5717, -0.2184] | -0.42 |
| | | Event recall | OpEntropy - TS-JEPA | 32 | 0.6528 | 0.7351 | 0.0000 [-0.0436, 0.0676] | -0.06 |
| | | Event recall | Reynolds - TS-JEPA | 32 | 0.8444 | 0.8444 | 0.0000 [-0.0586, 0.0554] | 0.00 |
| | | Event recall | PotentialFlow - TS-JEPA | 32 | 0.2770 | 0.4021 | 0.0000 [-0.1508, 0.0344] | -0.37 |
| | | Event recall | OpEntropy - TimesNet | 32 | 0.0003 | 0.0013 | -0.5167 [-0.6028, -0.2453] | -0.50 |
| | | Event recall | Reynolds - TimesNet | 32 | 0.0001 | 0.0013 | -0.4462 [-0.6063, -0.2734] | -0.64 |
| | | Event recall | PotentialFlow - TimesNet | 32 | $<$0.0001 | 0.0013 | -0.5778 [-0.6447, -0.3384] | -0.64 |
| | | Event recall | TimesNet - TranAD | 32 | 0.1095 | 0.2149 | 0.0000 [-0.0082, 0.1126] | 0.64 |
| | | Detection delay (steps) | TS-JEPA - TimesNet | 13 | 0.0681 | 0.1703 | 45.6000 [-0.1247, 136.8621] | 0.54 |
| | | Detection delay (steps) | TS-JEPA - TranAD | 13 | 0.1465 | 0.2523 | 58.8000 [-469.1299, 126.9832] | 0.69 |
| | | Detection delay (steps) | OpEntropy - TS-JEPA | 17 | 0.6441 | 0.7351 | 1.1667 [-37.9789, 25.0056] | 0.06 |
| | | Detection delay (steps) | Reynolds - TS-JEPA | 17 | 0.0980 | 0.2100 | 16.4667 [6.3093, 103.6322] | 0.38 |
| | | Detection delay (steps) | PotentialFlow - TS-JEPA | 14 | 0.6698 | 0.7351 | -7.6306 [-49.2915, 48.7951] | -0.14 |
| | | Detection delay (steps) | OpEntropy - TimesNet | 14 | 0.0464 | 0.1228 | 57.9500 [-1.1657, 109.9677] | 0.54 |
| | | Detection delay (steps) | Reynolds - TimesNet | 14 | 0.0353 | 0.1058 | 38.7815 [23.5277, 215.7128] | 0.57 |
| | | Detection delay (steps) | PotentialFlow - TimesNet | 13 | 0.1099 | 0.2149 | 62.5000 [-0.4251, 105.6324] | 0.50 |
| | | Detection delay (steps) | TimesNet - TranAD | 26 | 0.1626 | 0.2523 | 0.0000 [-300.2782, 3.3327] | -0.29 |

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